Data 260

Homework 6

Name: Kshitija Shinde
SJSU ID: 019146491

Value	**Definition**
SID4	6491
PORT_BASE	8000 + (6491 mod 900) = 8000 + 191 = 8191
PREFIX	"s" + 6491 = s6491
SEED	6491
VERIFY_SEED	260000 + 6491 = 266491
DOMAIN_ID	6491 mod 8 = 3
Assigned Domain	Grocery supply and recall notices

Hardware: MacBook Air, 13-inch, M4, 2025, Apple M4 chip, 16GB RAM, macOS Tahoe 26.5.2

Local model used: qwen3:8b via Ollama

Tagged commit hash: 32f75f59209bb2a2fd1bf8f19b96b522f4cd64b7

Repository: https://github.com/Kshitija-Shinde9/data260-6491

Part 1 - MongoDB Document API

This homework builds on HW5. Add a MongoDB-backed document view of your domain.

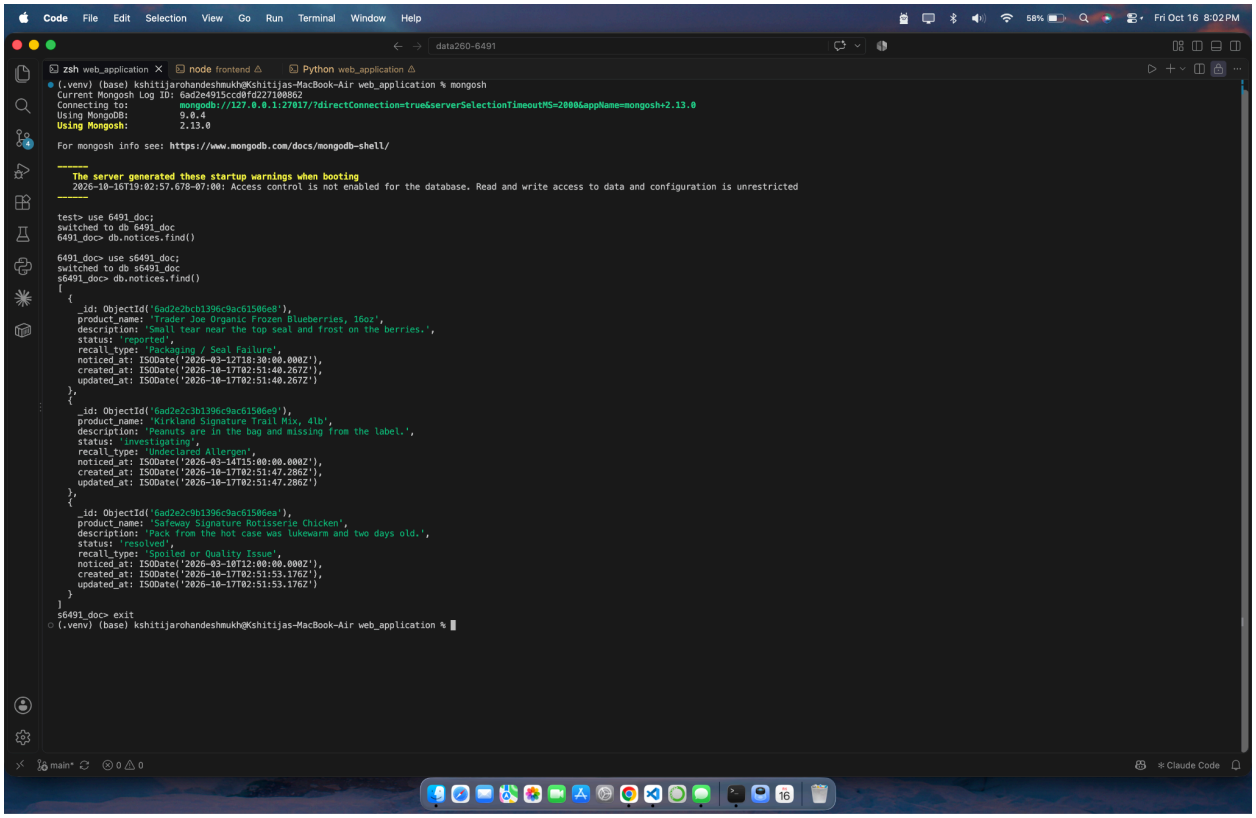
I. Database

- Create the MongoDB database _doc with a collection for a documentoriented representation of your domain (e.g., for a task-management-style setup this would be a "tasks" collection - map the fields below to your own domain entity instead).
- Include at least: one required short text field (max 100 characters), one optional description field, two constrained categorical fields (e.g., a status-style enum with a default, and a category-style field with a fixed set of options), one required date/time field, and automatic created/updated timestamps.
- The schema must include proper validation rules such as required fields, string length limits, and enum constraints.

```
DB_NAME = "s6491_doc"
COLLECTION_NAME = "notices"

NOTICE_SCHEMA = {
   "bsonType": "object",
   "required": [
       "product_name",
       "status",
       "recall_type",
       "noticed_at",
       "created_at",
       "updated_at",
   ],
   "properties": {
       "product_name": {"bsonType": "string", "minLength": 1,
"maxLength": 100},
       "description": {"bsonType": ["string", "null"], "maxLength":
2000},
       "status": {"enum": ["reported", "investigating", "resolved"]},
       "recall_type": {
           "enum": [
               "Packaging / Seal Failure",
               "Contamination",
```

```
                "Undeclared Allergen",
                "Spoiled or Quality Issue",
            ]
        },
        "noticed_at": {"bsonType": "date"},
        "created_at": {"bsonType": "date"},
        "updated_at": {"bsonType": "date"},
    },
}
```



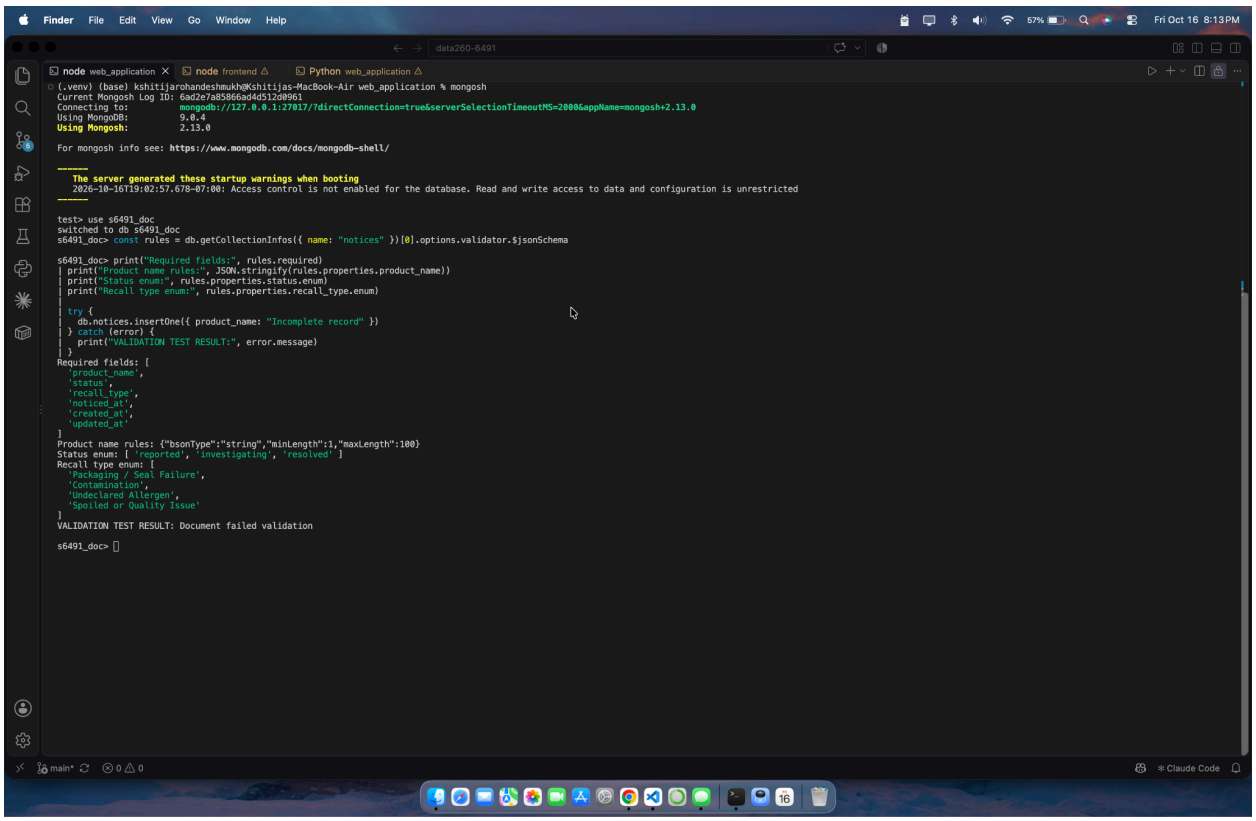
```
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % mongosh
Current Mongosh Log ID: 6ad2e4915ccd0fd227100862
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.13.0
Using MongoDB:          9.0.4
Using Mongosh:          2.13.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

------
   The server generated these startup warnings when booting
   2026-10-16T19:02:57.678-07:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
------

test> use 6491_doc;
switched to db 6491_doc
6491_doc> db.notices.find()

6491_doc> use s6491_doc;
switched to db s6491_doc
s6491_doc> db.notices.find()
[
  {
    _id: ObjectId('6ad2e2bcb1396c9ac61506e8'),
    product_name: 'Trader Joe Organic Frozen Blueberries, 16oz',
    description: 'Small tear near the top seal and frost on the berries.',
    status: 'reported',
    recall_type: 'Packaging / Seal Failure',
    noticed_at: ISODate('2026-03-12T18:30:00.000Z'),
    created_at: ISODate('2026-10-17T02:51:40.267Z'),
    updated_at: ISODate('2026-10-17T02:51:40.267Z')
  },
  {
    _id: ObjectId('6ad2e2c3b1396c9ac61506e9'),
    product_name: 'Kirkland Signature Trail Mix, 4lb',
    description: 'Peanuts are in the bag and missing from the label.',
    status: 'investigating',
    recall_type: 'Undeclared Allergen',
    noticed_at: ISODate('2026-03-14T15:00:00.000Z'),
    created_at: ISODate('2026-10-17T02:51:47.286Z'),
    updated_at: ISODate('2026-10-17T02:51:47.286Z')
  },
  {
    _id: ObjectId('6ad2e2c9b1396c9ac61506ea'),
    product_name: 'Safeway Signature Rotisserie Chicken',
    description: 'Pack from the hot case was lukewarm and two days old.',
    status: 'resolved',
    recall_type: 'Spoiled or Quality Issue',
    noticed_at: ISODate('2026-03-10T12:00:00.000Z'),
    created_at: ISODate('2026-10-17T02:51:53.176Z'),
    updated_at: ISODate('2026-10-17T02:51:53.176Z')
  }
]
s6491_doc> exit
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application %
```



```
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Current Mongosh Log ID: 6ad2e7a85866ad4d512d0961
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.13.0
Using MongoDB:          9.0.4
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------
   The server generated these startup warnings when booting
   2026-10-16T19:02:57.678-07:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
------

test> use s6491_doc
switched to db s6491_doc
s6491_doc> const rules = db.getCollectionInfos({ name: "notices" })[0].options.validator.$jsonSchema

s6491_doc> print("Required fields:", rules.required)
| print("Product name rules:", JSON.stringify(rules.properties.product_name))
| print("Status enum:", rules.properties.status.enum)
| print("Recall type enum:", rules.properties.recall_type.enum)
|
| try {
|   db.notices.insertOne({ product_name: "Incomplete record" })
| } catch (error) {
|   print("VALIDATION TEST RESULT:", error.message)
| }
Required fields: [
  'product_name',
  'status',
  'recall_type',
  'noticed_at',
  'created_at',
  'updated_at'
]
Product name rules: {"bsonType":"string","minLength":1,"maxLength":100}
Status enum: [ 'reported', 'investigating', 'resolved' ]
Recall type enum: [
  'Packaging / Seal Failure',
  'Contamination',
  'Undeclared Allergen',
  'Spoiled or Quality Issue'
]
VALIDATION TEST RESULT: Document failed validation

s6491_doc>
```
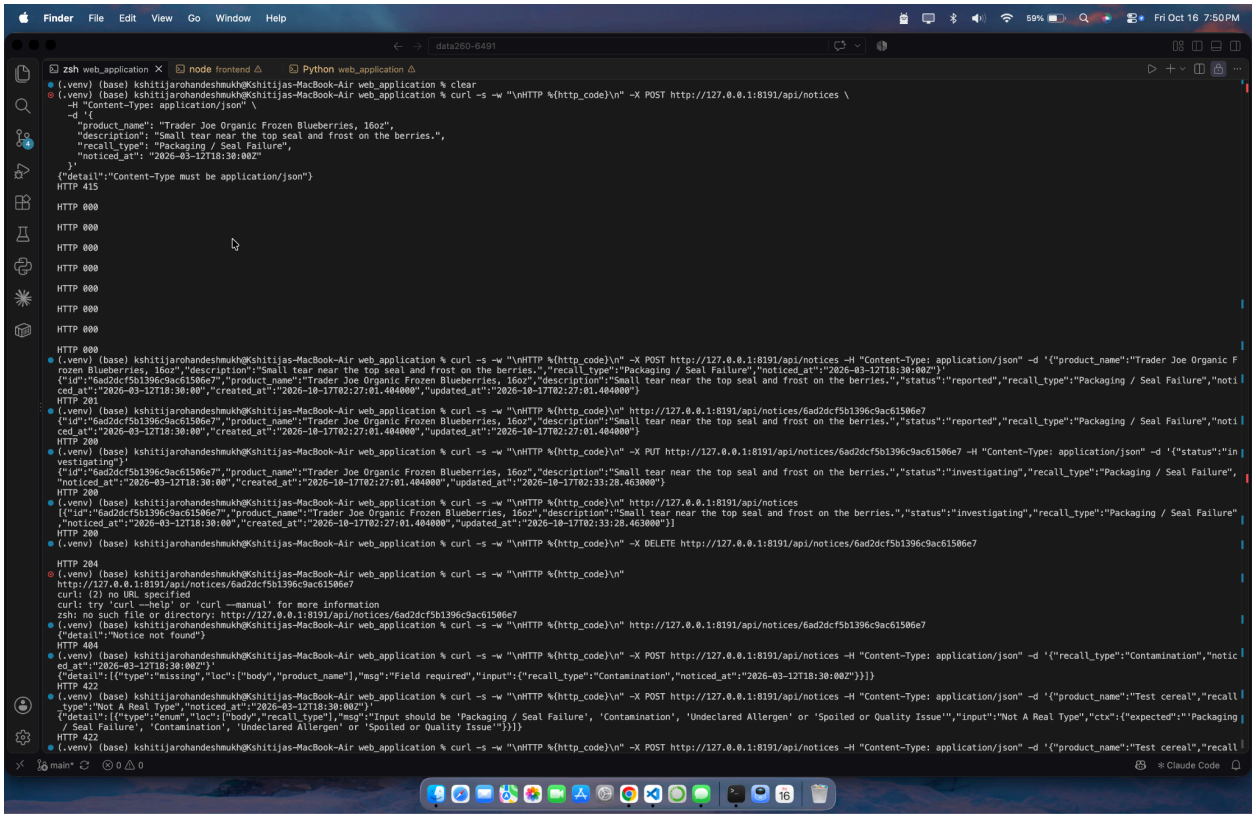

II. API Implementation

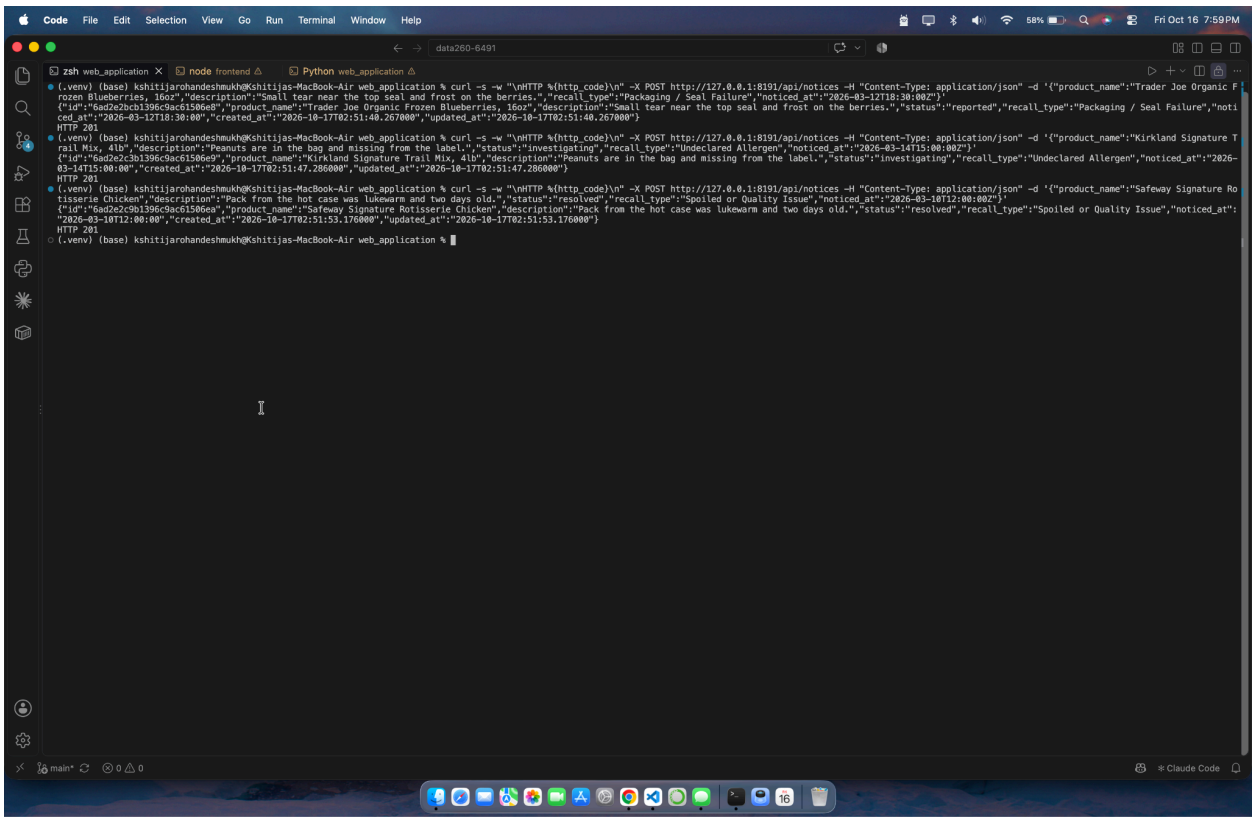
- Implement a REST API using FastAPI with complete CRUD operations: create (POST), retrieve all (GET), get a single record by ID (GET /:id), update (PUT /:id), delete (DELETE /:id).
- Include proper error handling with appropriate HTTP status codes for scenarios like invalid data, resource not found, and server errors.
- Use middleware for CORS and JSON parsing for request bodies. Structure your code with separate files for server configuration, PyMongo database connection, data models/schemas (Pydantic), and route handlers.

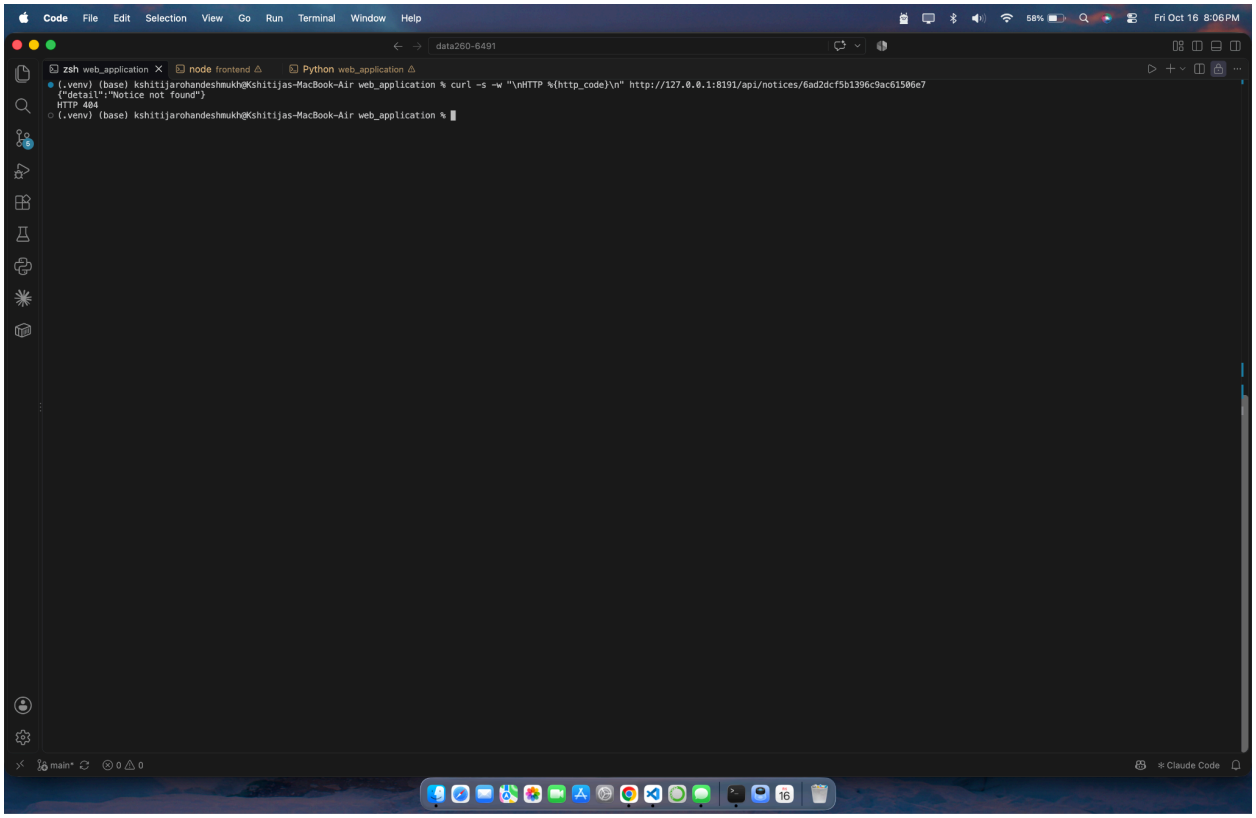
```
app.add_middleware(
   CORSMiddleware,
   allow_origins=["http://localhost:5173"],
   allow_credentials=True,
   allow_methods=["*"],
   allow_headers=["*"],
)

app.add_middleware(JsonBodyMiddleware)

app.include_router(auth_router)
app.include_router(api_auth_router)
app.include_router(suppliers_router)
app.include_router(recalls_router)
app.include_router(bench_router)
app.include_router(notices_router)
```

```
router = APIRouter(prefix="/api/notices", tags=["notices"])

@router.post("", response_model=NoticeOut, status_code=201)
def create_notice(body: NoticeCreate): ...

@router.get("", response_model=List[NoticeOut])
def list_notices(): ...

@router.get("/{notice_id}", response_model=NoticeOut)
def get_notice(notice_id: str): ...
```

```
@router.put("/{notice_id}", response_model=NoticeOut)
def update_notice(notice_id: str, body: NoticeUpdate): ...

@router.delete("/{notice_id}", status_code=204)
def delete_notice(notice_id: str): ...
```


Finder File Edit View Go Window Help
Fri Oct 16 7:50 PM
zsh web_application
node frontend
Python web_application
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % clear
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" -X POST http://127.0.0.1:8191/api/notices \
-H "Content-Type: application/json" \
-d '{
"product_name": "Trader Joe Organic Frozen Blueberries, 16oz",
"description": "Small tear near the top seal and frost on the berries.",
"recall_type": "Packaging / Seal Failure",
"noticed_at": "2026-03-12T18:30:00Z"
}'
{"detail":"Content-Type must be application/json"}
HTTP 415
HTTP 000
HTTP 000
HTTP 000
HTTP 000
HTTP 000
HTTP 000
HTTP 000
HTTP 000
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" -X POST http://127.0.0.1:8191/api/notices -H "Content-Type: application/json" -d '{"product_name":"Trader Joe Organic Frozen Blueberries, 16oz","description":"Small tear near the top seal and frost on the berries.","recall_type":"Packaging / Seal Failure","noticed_at":"2026-03-12T18:30:00Z"}'
{"id":"6ad2dcf5b1396c9ac61506e7","product_name":"Trader Joe Organic Frozen Blueberries, 16oz","description":"Small tear near the top seal and frost on the berries.","status":"reported","recall_type":"Packaging / Seal Failure","noticed_at":"2026-03-12T18:30:00","created_at":"2026-10-17T02:27:01.404000","updated_at":"2026-10-17T02:27:01.404000"}
HTTP 201
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" http://127.0.0.1:8191/api/notices/6ad2dcf5b1396c9ac61506e7
{"id":"6ad2dcf5b1396c9ac61506e7","product_name":"Trader Joe Organic Frozen Blueberries, 16oz","description":"Small tear near the top seal and frost on the berries.","status":"reported","recall_type":"Packaging / Seal Failure","noticed_at":"2026-03-12T18:30:00","created_at":"2026-10-17T02:27:01.404000","updated_at":"2026-10-17T02:27:01.404000"}
HTTP 200
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" -X PUT http://127.0.0.1:8191/api/notices/6ad2dcf5b1396c9ac61506e7 -H "Content-Type: application/json" -d '{"status":"investigating"}'
{"id":"6ad2dcf5b1396c9ac61506e7","product_name":"Trader Joe Organic Frozen Blueberries, 16oz","description":"Small tear near the top seal and frost on the berries.","status":"investigating","recall_type":"Packaging / Seal Failure","noticed_at":"2026-03-12T18:30:00","created_at":"2026-10-17T02:27:01.404000","updated_at":"2026-10-17T02:33:28.463000"}
HTTP 200
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" http://127.0.0.1:8191/api/notices
[{"id":"6ad2dcf5b1396c9ac61506e7","product_name":"Trader Joe Organic Frozen Blueberries, 16oz","description":"Small tear near the top seal and frost on the berries.","status":"investigating","recall_type":"Packaging / Seal Failure","noticed_at":"2026-03-12T18:30:00","created_at":"2026-10-17T02:27:01.404000","updated_at":"2026-10-17T02:33:28.463000"}]
HTTP 200
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" -X DELETE http://127.0.0.1:8191/api/notices/6ad2dcf5b1396c9ac61506e7
HTTP 204
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n"
http://127.0.0.1:8191/api/notices/6ad2dcf5b1396c9ac61506e7
curl: (2) no URL specified
curl: try 'curl --help' or 'curl --manual' for more information
zsh: no such file or directory: http://127.0.0.1:8191/api/notices/6ad2dcf5b1396c9ac61506e7
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" http://127.0.0.1:8191/api/notices/6ad2dcf5b1396c9ac61506e7
{"detail":"Notice not found"}
HTTP 404
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" -X POST http://127.0.0.1:8191/api/notices -H "Content-Type: application/json" -d '{"recall_type":"Contamination","noticed_at":"2026-03-12T18:30:00Z"}'
{"detail":[{"type":"missing","loc":["body","product_name"],"msg":"Field required","input":{"recall_type":"Contamination","noticed_at":"2026-03-12T18:30:00Z"}}]}
HTTP 422
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" -X POST http://127.0.0.1:8191/api/notices -H "Content-Type: application/json" -d '{"product_name":"Test cereal","recall_type":"Not A Real Type","noticed_at":"2026-03-12T18:30:00Z"}'
{"detail":[{"type":"enum","loc":["body","recall_type"],"msg":"Input should be 'Packaging / Seal Failure', 'Contamination', 'Undeclared Allergen' or 'Spoiled or Quality Issue'","input":"Not A Real Type","ctx":{"expected":"'Packaging / Seal Failure', 'Contamination', 'Undeclared Allergen' or 'Spoiled or Quality Issue'"}}]}
HTTP 422
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" -X POST http://127.0.0.1:8191/api/notices -H "Content-Type: application/json" -d '{"product_name":"Test cereal","recall
main*
0 0
Claude Code


Code File Edit Selection View Go Run Terminal Window Help
Fri Oct 16 7:59 PM
zsh web_application
node frontend
Python web_application
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" -X POST http://127.0.0.1:8191/api/notices -H "Content-Type: application/json" -d '{"product_name":"Trader Joe Organic Frozen Blueberries, 16oz","description":"Small tear near the top seal and frost on the berries.","recall_type":"Packaging / Seal Failure","noticed_at":"2026-03-12T18:30:00Z"}'
{"id":"6ad2e2bcb1396c9ac61506e8","product_name":"Trader Joe Organic Frozen Blueberries, 16oz","description":"Small tear near the top seal and frost on the berries.","status":"reported","recall_type":"Packaging / Seal Failure","noticed_at":"2026-03-12T18:30:00","created_at":"2026-10-17T02:51:40.267000","updated_at":"2026-10-17T02:51:40.267000"}
HTTP 201
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" -X POST http://127.0.0.1:8191/api/notices -H "Content-Type: application/json" -d '{"product_name":"Kirkland Signature Trail Mix, 4lb","description":"Peanuts are in the bag and missing from the label.","status":"investigating","recall_type":"Undeclared Allergen","noticed_at":"2026-03-14T15:00:00Z"}'
{"id":"6ad2e2c3b1396c9ac61506e9","product_name":"Kirkland Signature Trail Mix, 4lb","description":"Peanuts are in the bag and missing from the label.","status":"investigating","recall_type":"Undeclared Allergen","noticed_at":"2026-03-14T15:00:00","created_at":"2026-10-17T02:51:47.286000","updated_at":"2026-10-17T02:51:47.286000"}
HTTP 201
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" -X POST http://127.0.0.1:8191/api/notices -H "Content-Type: application/json" -d '{"product_name":"Safeway Signature Rotisserie Chicken","description":"Pack from the hot case was lukewarm and two days old.","status":"resolved","recall_type":"Spoiled or Quality Issue","noticed_at":"2026-03-10T12:00:00Z"}'
{"id":"6ad2e2c9b1396c9ac61506ea","product_name":"Safeway Signature Rotisserie Chicken","description":"Pack from the hot case was lukewarm and two days old.","status":"resolved","recall_type":"Spoiled or Quality Issue","noticed_at":"2026-03-10T12:00:00","created_at":"2026-10-17T02:51:53.176000","updated_at":"2026-10-17T02:51:53.176000"}
HTTP 201
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application %
main*
0 0
Claude Code


Code File Edit Selection View Go Run Terminal Window Help
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zsh web_application
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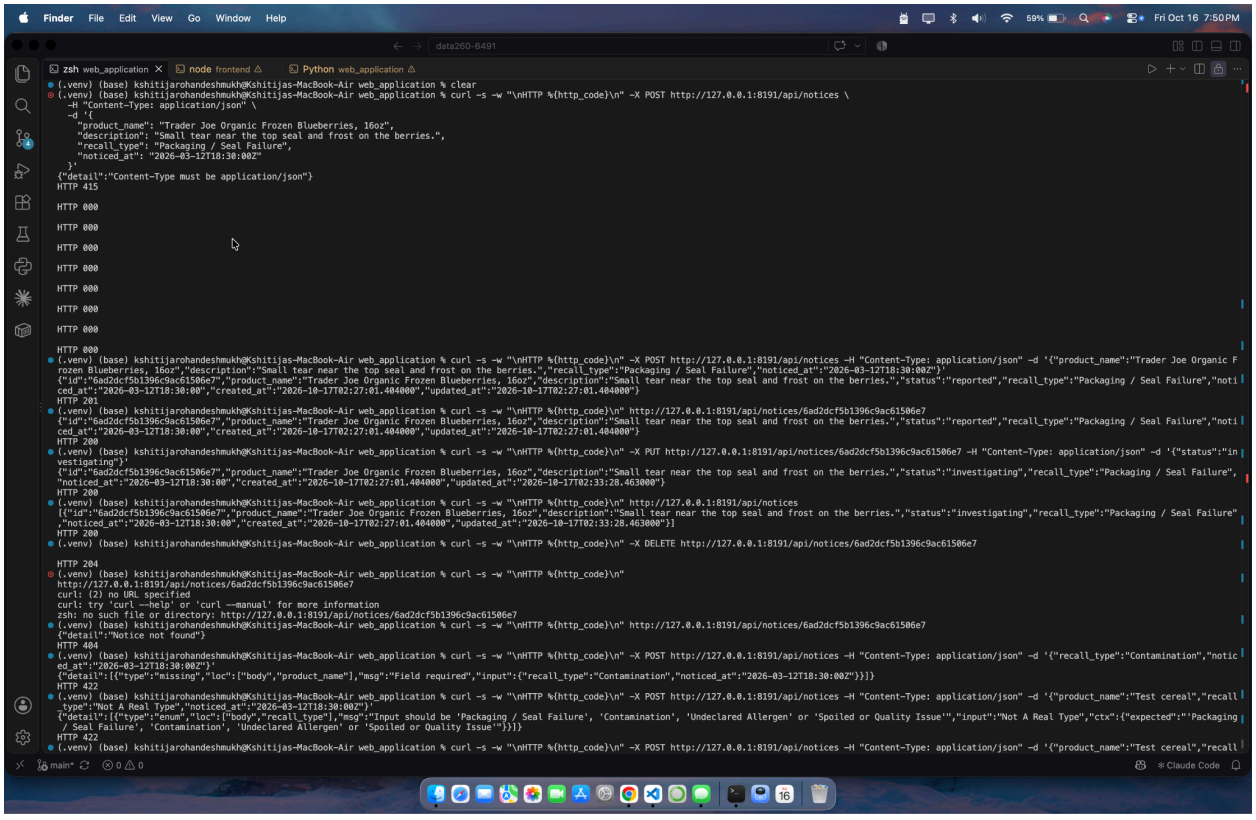
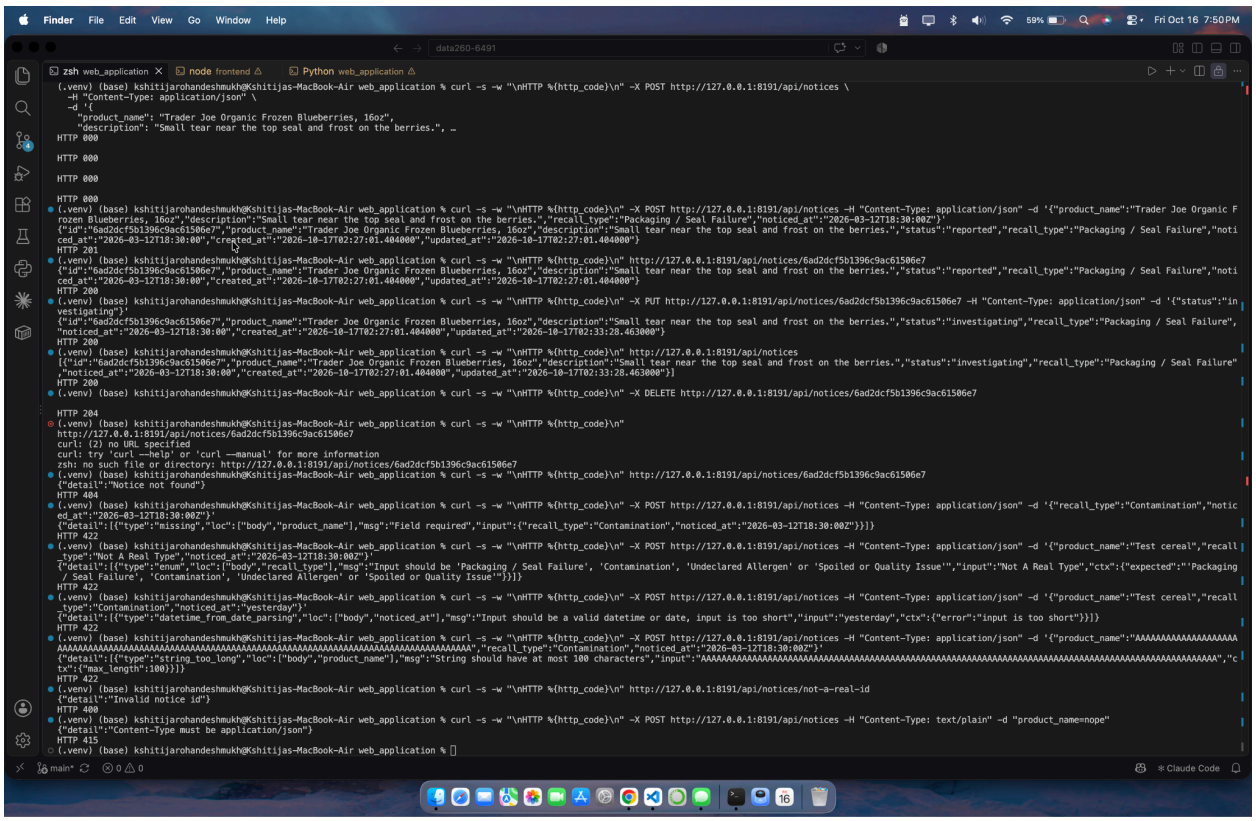
{"detail":"Notice not found"}
HTTP 404
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application %
main*
0 0
Claude Code

III. Data Validation

- Implement validation using Pydantic schemas to enforce data integrity including required fields, string lengths, enum values, and proper date formats.
- Ensure that validation errors return clear error messages with appropriate HTTP status codes.
- Test validation by attempting to create records with invalid data such as missing required fields, invalid enum values, and incorrect data types.

```
class NoticeCreate(BaseModel):
    product_name: str = Field(min_length=1, max_length=100)
    description: Optional[str] = Field(default=None, max_length=2000)
    status: NoticeStatus = NoticeStatus.reported
    recall_type: RecallType
    noticed_at: datetime
```

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-H "Content-Type: application/json" \
-d '{
"product_name": "Trader Joe Organic Frozen Blueberries, 16oz",
"description": "Small tear near the top seal and frost on the berries.", …
HTTP 000
HTTP 000
HTTP 000
HTTP 000
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" -X POST http://127.0.0.1:8191/api/notices -H "Content-Type: application/json" -d '{"product_name":"Trader Joe Organic Frozen Blueberries, 16oz","description":"Small tear near the top seal and frost on the berries.","recall_type":"Packaging / Seal Failure","noticed_at":"2026-03-12T18:30:00Z"}'
{"id":"6ad2dcf5b1396c9ac61506e7","product_name":"Trader Joe Organic Frozen Blueberries, 16oz","description":"Small tear near the top seal and frost on the berries.","status":"reported","recall_type":"Packaging / Seal Failure","noticed_at":"2026-03-12T18:30:00","created_at":"2026-10-17T02:27:01.404000","updated_at":"2026-10-17T02:27:01.404000"}
HTTP 201
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" http://127.0.0.1:8191/api/notices/6ad2dcf5b1396c9ac61506e7
{"id":"6ad2dcf5b1396c9ac61506e7","product_name":"Trader Joe Organic Frozen Blueberries, 16oz","description":"Small tear near the top seal and frost on the berries.","status":"reported","recall_type":"Packaging / Seal Failure","noticed_at":"2026-03-12T18:30:00","created_at":"2026-10-17T02:27:01.404000","updated_at":"2026-10-17T02:27:01.404000"}
HTTP 200
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" -X PUT http://127.0.0.1:8191/api/notices/6ad2dcf5b1396c9ac61506e7 -H "Content-Type: application/json" -d '{"status":"investigating"}'
{"id":"6ad2dcf5b1396c9ac61506e7","product_name":"Trader Joe Organic Frozen Blueberries, 16oz","description":"Small tear near the top seal and frost on the berries.","status":"investigating","recall_type":"Packaging / Seal Failure","noticed_at":"2026-03-12T18:30:00","created_at":"2026-10-17T02:27:01.404000","updated_at":"2026-10-17T02:33:28.463000"}
HTTP 200
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" http://127.0.0.1:8191/api/notices
[{"id":"6ad2dcf5b1396c9ac61506e7","product_name":"Trader Joe Organic Frozen Blueberries, 16oz","description":"Small tear near the top seal and frost on the berries.","status":"investigating","recall_type":"Packaging / Seal Failure","noticed_at":"2026-03-12T18:30:00","created_at":"2026-10-17T02:27:01.404000","updated_at":"2026-10-17T02:33:28.463000"}]
HTTP 200
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" -X DELETE http://127.0.0.1:8191/api/notices/6ad2dcf5b1396c9ac61506e7
HTTP 204
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n"
http://127.0.0.1:8191/api/notices/6ad2dcf5b1396c9ac61506e7
curl: (2) no URL specified
curl: try 'curl --help' or 'curl --manual' for more information
zsh: no such file or directory: http://127.0.0.1:8191/api/notices/6ad2dcf5b1396c9ac61506e7
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" http://127.0.0.1:8191/api/notices/6ad2dcf5b1396c9ac61506e7
{"detail":"Notice not found"}
HTTP 404
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" -X POST http://127.0.0.1:8191/api/notices -H "Content-Type: application/json" -d '{"recall_type":"Contamination","noticed_at":"2026-03-12T18:30:00Z"}'
{"detail":[{"type":"missing","loc":["body","product_name"],"msg":"Field required","input":{"recall_type":"Contamination","noticed_at":"2026-03-12T18:30:00Z"}}]}
HTTP 422
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" -X POST http://127.0.0.1:8191/api/notices -H "Content-Type: application/json" -d '{"product_name":"Test cereal","recall_type":"Not A Real Type","noticed_at":"2026-03-12T18:30:00Z"}'
{"detail":[{"type":"enum","loc":["body","recall_type"],"msg":"Input should be 'Packaging / Seal Failure', 'Contamination', 'Undeclared Allergen' or 'Spoiled or Quality Issue'","input":"Not A Real Type","ctx":{"expected":"'Packaging / Seal Failure', 'Contamination', 'Undeclared Allergen' or 'Spoiled or Quality Issue'"}}]}
HTTP 422
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" -X POST http://127.0.0.1:8191/api/notices -H "Content-Type: application/json" -d '{"product_name":"Test cereal","recall_type":"Contamination","noticed_at":"yesterday"}'
{"detail":[{"type":"datetime_from_date_parsing","loc":["body","noticed_at"],"msg":"Input should be a valid datetime or date, input is too short","input":"yesterday","ctx":{"error":"input is too short"}}]}
HTTP 422
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % curl -s -w "\nHTTP %{http_code}\n" -X POST http://127.0.0.1:8191/api/notices -H "Content-Type: application/json" -d '{"product_name":"AAAA

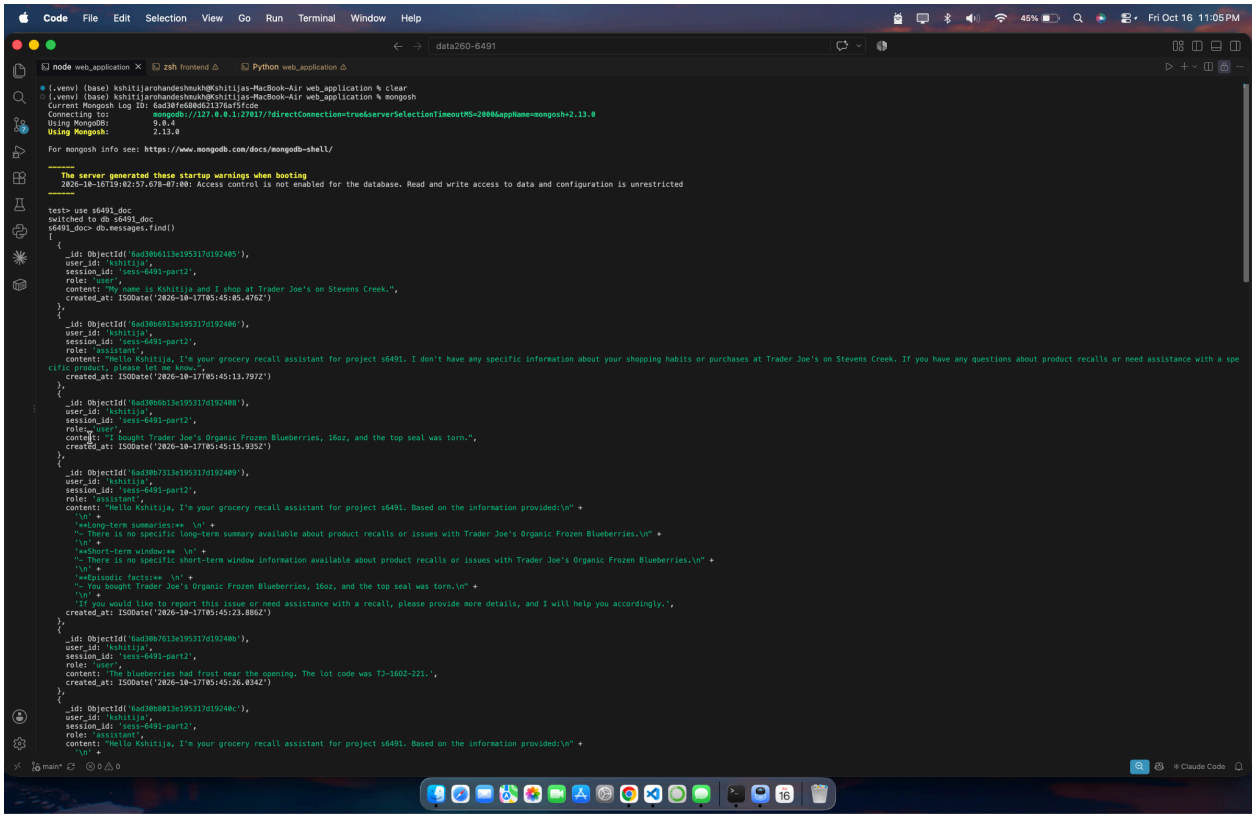
Part 2 - Short-, Long-, and Episodic Memory

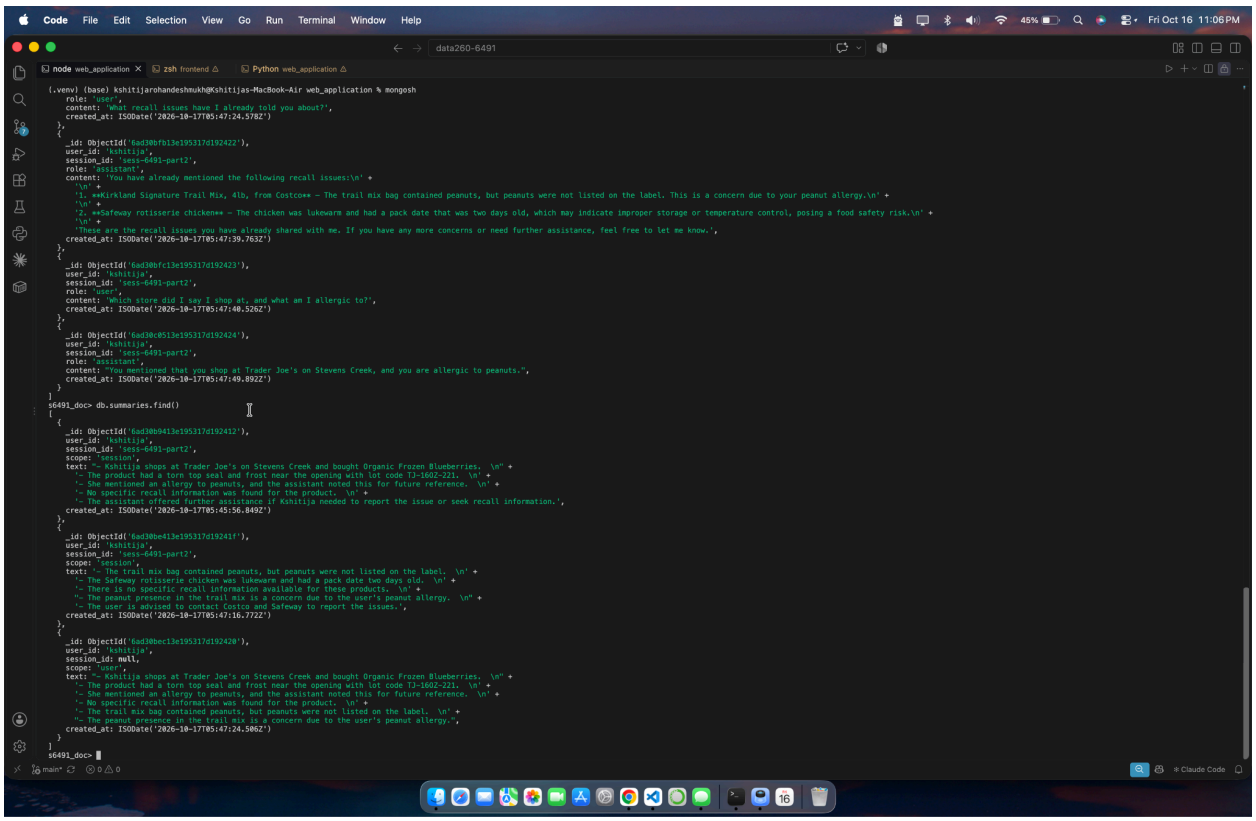
Use the same MongoDB database with new collections: messages, summaries, episodes. Keep your Part 1 collection unchanged - you are only reusing the same database/connection.

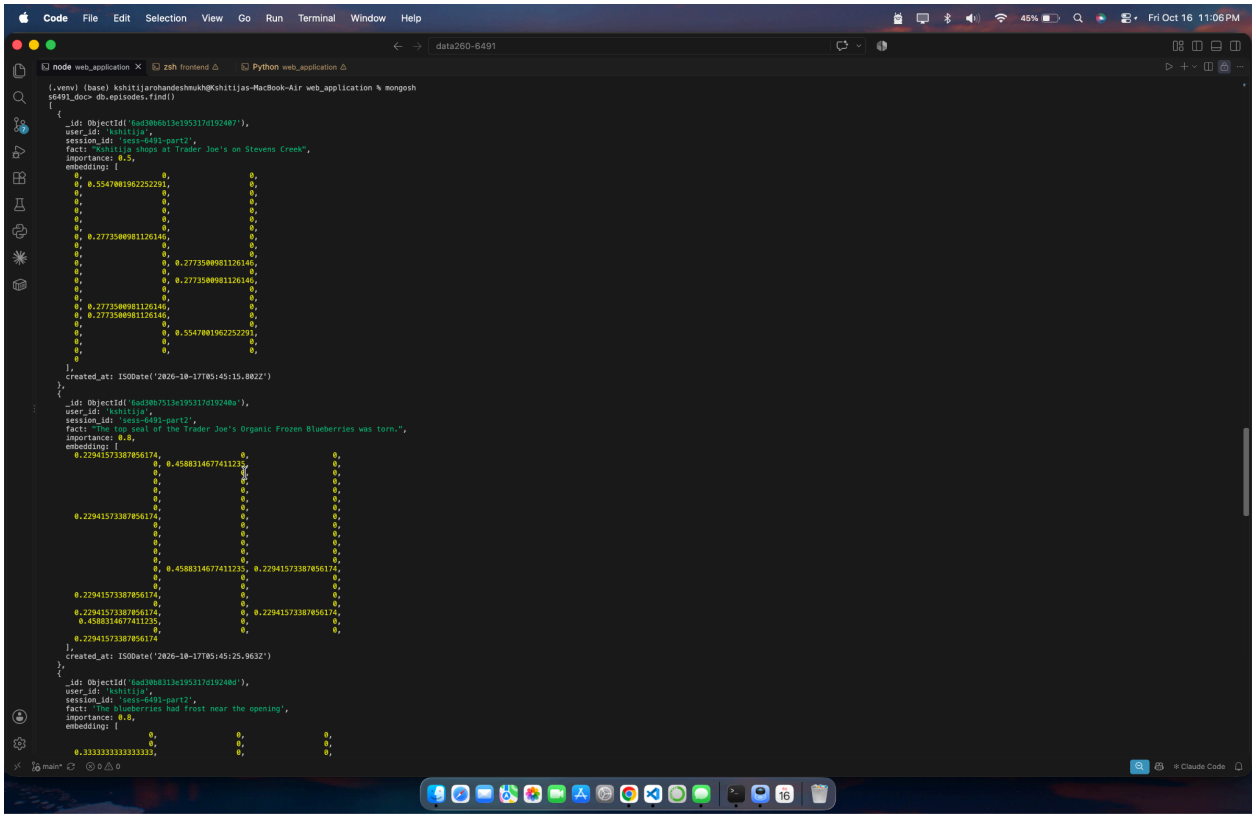
MongoDB collections (reuse the same DB)

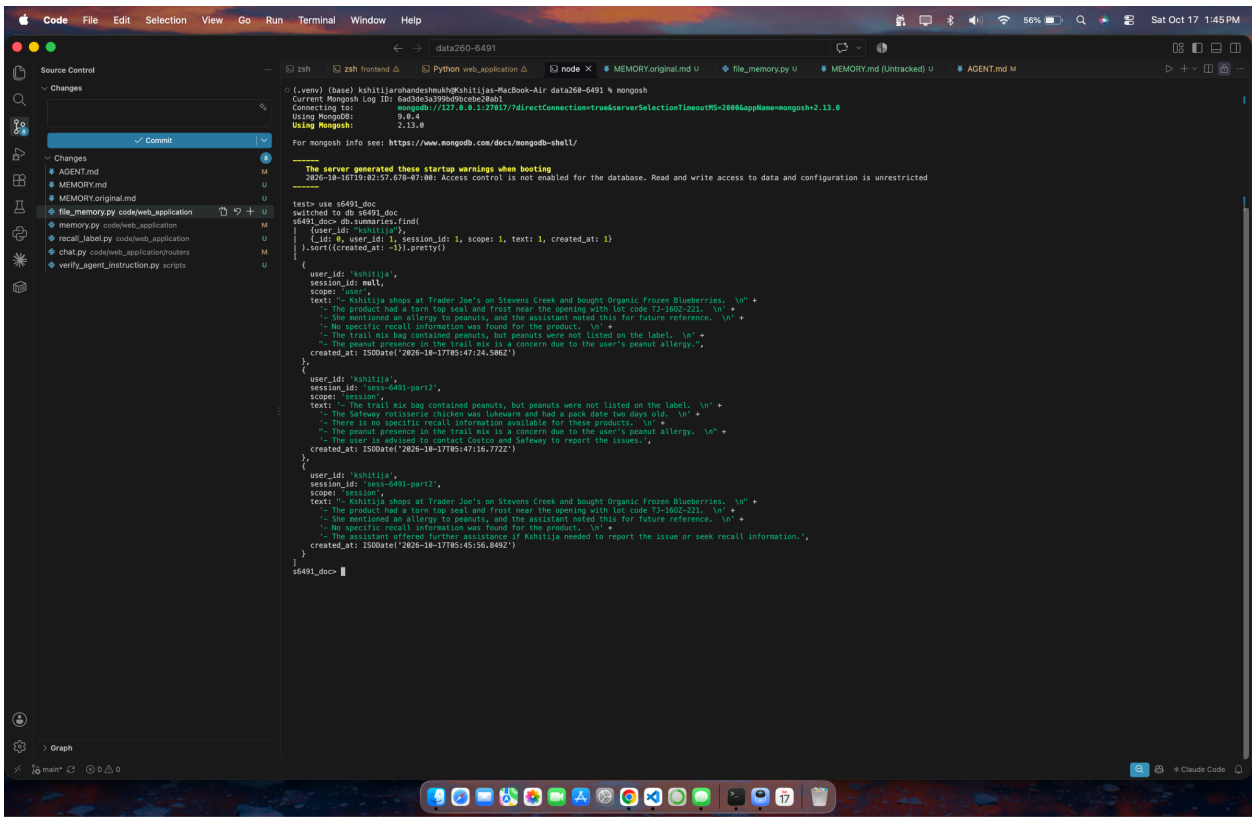
Create new collections: (3 points)
Messages
Fields: user_id, session_id, role (user/assistant), content, created_at.
Summaries
Fields: user_id, session_id (nullable for lifetime), scope ('session' | 'user'), text, created_at.
Episodes
Fields: user_id, session_id, fact (short text), importance (0..1), embedding (array of floats), created_at.

```
def ensure_memory_collections() -> None:
    _ensure_collection("messages", MESSAGE_SCHEMA)
    _ensure_collection("summaries", SUMMARY_SCHEMA)
    _ensure_collection("episodes", EPISODE_SCHEMA)
```


Code File Edit Selection View Go Run Terminal Window Help
Fri Oct 16 11:05 PM
data260-6491
node web_application
zsh frontend
Python web_application
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % clear
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % mongosh
Current Mongosh Log ID: 6ad30fe680d621376af5fcde
Connecting to: mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.13.0
Using MongoDB: 9.0.4
Using Mongosh: 2.13.0
For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/
The server generated these startup warnings when booting
2026-10-16T19:02:57.678-07:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
test> use s6491_doc
switched to db s6491_doc
s6491_doc> db.messages.find()
'My name is Kshitija and I shop at Trader Joe's on Stevens Creek.'
'Hello Kshitija, I'm your grocery recall assistant for project s6491. I don't have any specific information about your shopping habits or purchases at Trader Joe's on Stevens Creek. If you have any questions about product recalls or need assistance with a specific product, please let me know.'
'I bought Trader Joe's Organic Frozen Blueberries, 16oz, and the top seal was torn.'
'Hello Kshitija, I'm your grocery recall assistant for project s6491. Based on the information provided:\n'
'The blueberries had frost near the opening. The lot code was TJ-16OZ-221.'
'Hello Kshitija, I'm your grocery recall assistant for project s6491. Based on the information provided:\n'


Code File Edit Selection View Go Run Terminal Window Help
Fri Oct 16 11:06 PM
data260-6491
node web_application
zsh frontend
Python web_application
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % mongosh
'What recall issues have I already told you about?'
'You have already mentioned the following recall issues:\n'
'1. **Kirkland Signature Trail Mix, 4lb, from Costco** - The trail mix bag contained peanuts, but peanuts were not listed on the label. This is a concern due to your peanut allergy.\n'
'2. **Safeway rotisserie chicken** - The chicken was lukewarm and had a pack date that was two days old, which may indicate improper storage or temperature control, posing a food safety risk.\n'
'These are the recall issues you have already shared with me. If you have any more concerns or need further assistance, feel free to let me know.'
'Which store did I say I shop at, and what am I allergic to?'
'You mentioned that you shop at Trader Joe's on Stevens Creek, and you are allergic to peanuts.'
s6491_doc> db.summaries.find()
'~ Kshitija shops at Trader Joe's on Stevens Creek and bought Organic Frozen Blueberries.\n'
'- The product had a torn top seal and frost near the opening with lot code TJ-16O2-221.\n'
'- She mentioned an allergy to peanuts, and the assistant noted this for future reference.\n'
'- No specific recall information was found for the product.\n'
'- The assistant offered further assistance if Kshitija needed to report the issue or seek recall information.'
'~ The trail mix bag contained peanuts, but peanuts were not listed on the label.\n'
'- The Safeway rotisserie chicken was lukewarm and had a pack date two days old.\n'
'- There is no specific recall information available for these products.\n'
'- The peanut presence in the trail mix is a concern due to the user's peanut allergy.\n'
'- The user is advised to contact Costco and Safeway to report the issues.'
'~ Kshitija shops at Trader Joe's on Stevens Creek and bought Organic Frozen Blueberries.\n'
'- The product had a torn top seal and frost near the opening with lot code TJ-16O2-221.\n'
'- She mentioned an allergy to peanuts, and the assistant noted this for future reference.\n'
'- No specific recall information was found for the product.\n'
'- The trail mix bag contained peanuts, but peanuts were not listed on the label.\n'
'- The peanut presence in the trail mix is a concern due to the user's peanut allergy.'
s6491_doc>


(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % mongosh
s6491_doc> db.episodes.find()
_id: ObjectId('6ad30b6b13e195317d19240f'),
user_id: 'kshitija',
session_id: 'sess-6491-part2',
fact: "Kshitija shops at Trader Joe's on Stevens Creek",
importance: 0.5,
_id: ObjectId('6ad30b7513e195317d19240a'),
user_id: 'kshitija',
session_id: 'sess-6491-part2',
fact: "The top seal of the Trader Joe's Organic Frozen Blueberries was torn.",
importance: 0.8,
_id: ObjectId('6ad30b8313e195317d19240d'),
user_id: 'kshitija',
session_id: 'sess-6491-part2',
fact: 'The blueberries had frost near the opening',
importance: 0.8,


(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air data260-6491 % mongosh
Current Mongosh Log ID: 6ad3de3a399bd9bcebe20ab1
Connecting to: mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.13.0
Using MongoDB: 9.0.4
Using Mongosh: 2.13.0
For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/
The server generated these startup warnings when booting
2026-10-16T19:02:57.678-07:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
test> use s6491_doc
switched to db s6491_doc
s6491_doc> db.summaries.find(
{user_id: 'kshitija'},
{_id: 0, user_id: 1, session_id: 1, scope: 1, text: 1, created_at: 1}
).sort({created_at: -1}).pretty()
user_id: 'kshitija',
session_id: null,
scope: 'user',
text: '- Kshitija shops at Trader Joe's on Stevens Creek and bought Organic Frozen Blueberries. \n' +
'- The product had a torn top seal and frost near the opening with lot code TJ-16OZ-221. \n' +
'- She mentioned an allergy to peanuts, and the assistant noted this for future reference. \n' +
'- No specific recall information was found for the product. \n' +
'- The trail mix bag contained peanuts, but peanuts were not listed on the label. \n' +
"- The peanut presence in the trail mix is a concern due to the user's peanut allergy.",
created_at: ISODate('2026-10-17T05:47:24.506Z')
user_id: 'kshitija',
session_id: 'sess-6491-part2',
scope: 'session',
text: '- The trail mix bag contained peanuts, but peanuts were not listed on the label. \n' +
'- The Safeway rotisserie chicken was lukewarm and had a pack date two days old. \n' +
'- There is no specific recall information available for these products. \n' +
"- The peanut presence in the trail mix is a concern due to the user's peanut allergy. \n" +
'- The user is advised to contact Costco and Safeway to report the issues.',
created_at: ISODate('2026-10-17T05:47:16.772Z')
user_id: 'kshitija',
session_id: 'sess-6491-part2',
scope: 'session',
text: '- Kshitija shops at Trader Joe's on Stevens Creek and bought Organic Frozen Blueberries. \n' +
'- The product had a torn top seal and frost near the opening with lot code TJ-16OZ-221. \n' +
'- She mentioned an allergy to peanuts, and the assistant noted this for future reference. \n' +
'- No specific recall information was found for the product. \n' +
'- The assistant offered further assistance if Kshitija needed to report the issue or seek recall information.',
created_at: ISODate('2026-10-17T05:45:56.849Z')
s6491_doc>

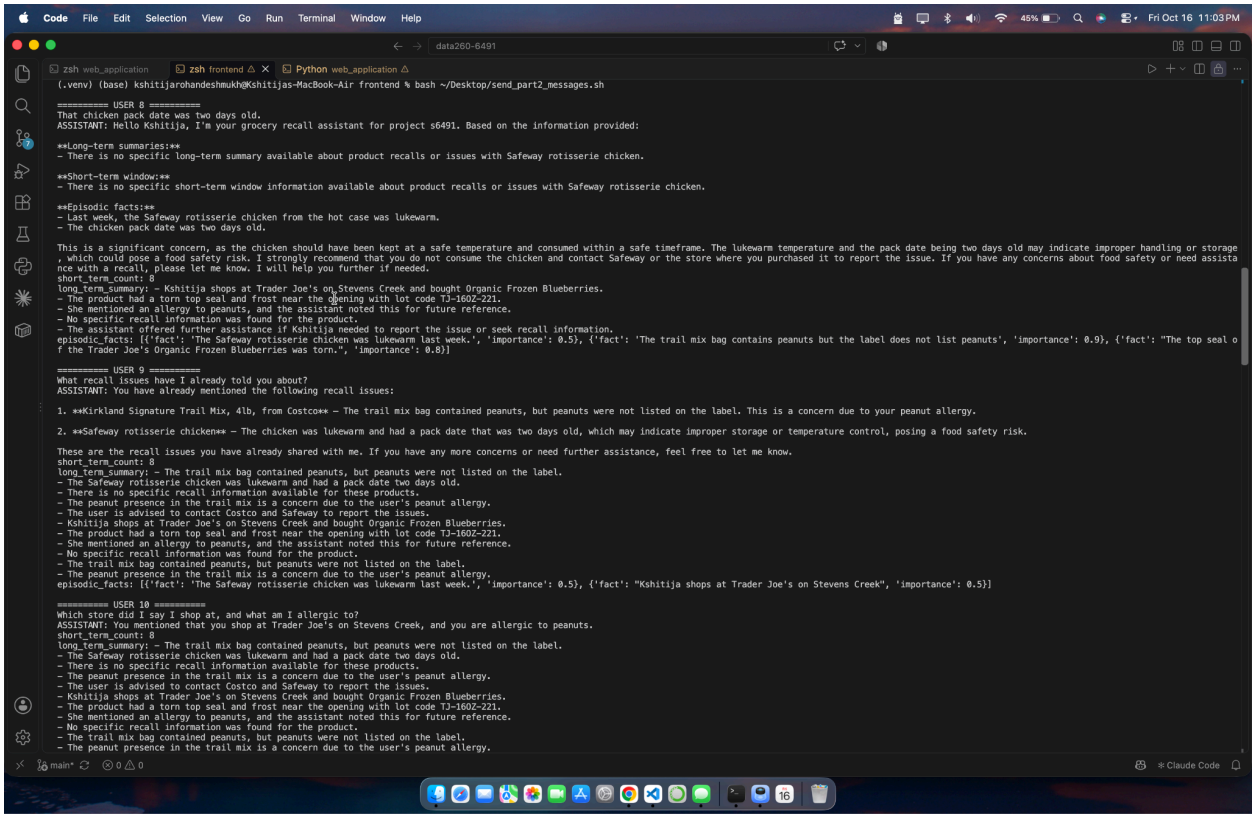
FastAPI Endpoints

1. POST /api/chat
   - Request: { user_id, session_id (optional), message }
   - Server steps: save the user message to messages; load the short-term window (last N messages, N = SHORT_TERM_N); load the latest session summary and lifetime summary from summaries; retrieve the top-k relevant episodic facts; compose the prompt (system primer + long-term summaries + short-term window + current message + episodic facts); call the model through your HW1 model adapter (local Ollama model); save the assistant reply to messages; every SUMMARIZE_EVERY_USER_MSGS user messages, generate/update a session summary, and occasionally condense session summaries into a lifetime summary.
   - Response: the assistant reply plus an object showing what was used - short-term message count/list, the long-term summary text (if any), and the episodic facts retrieved for this turn.

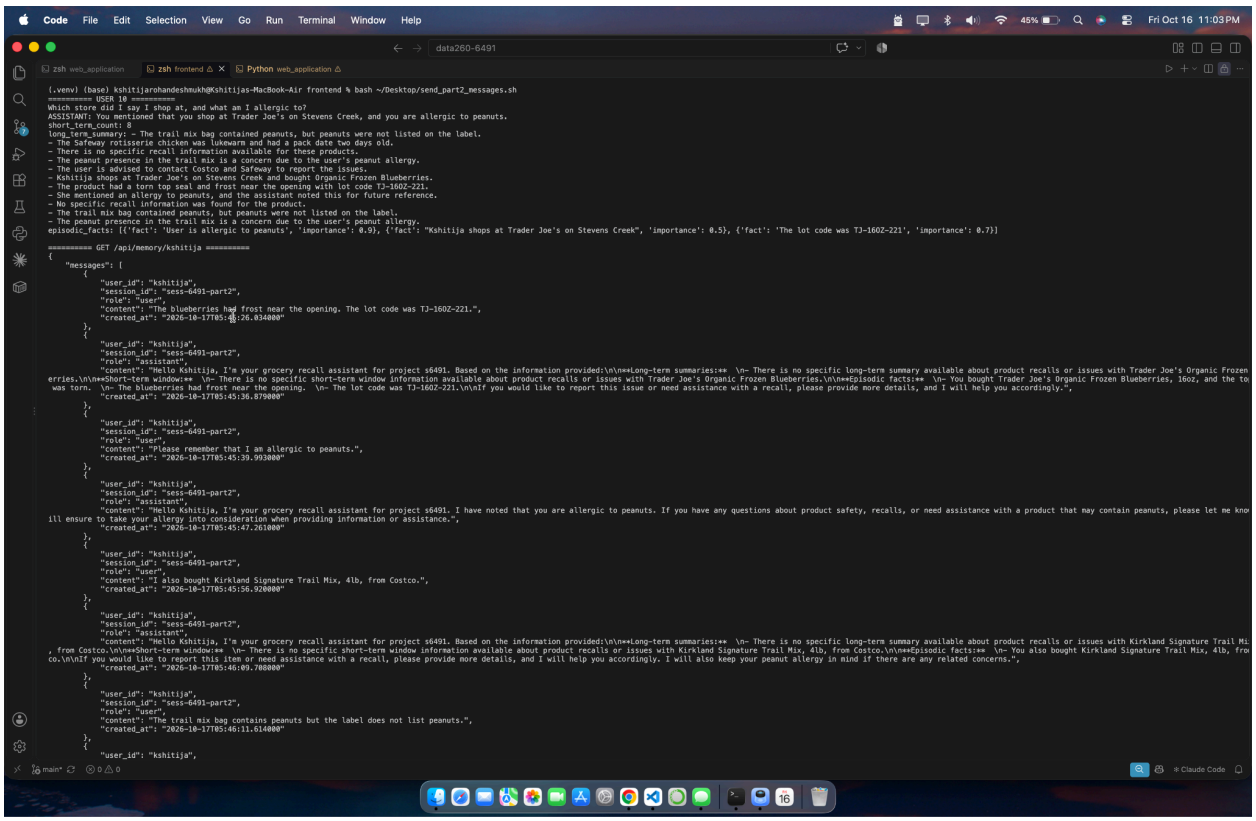
2. GET /api/memory/{user_id} - returns the last ~16 messages, latest session summary, latest lifetime summary, and last ~20 episodic facts.

3. GET /api/aggregate/{user_id} - returns daily message counts (grouped by date) and a small list of the most recent summaries.

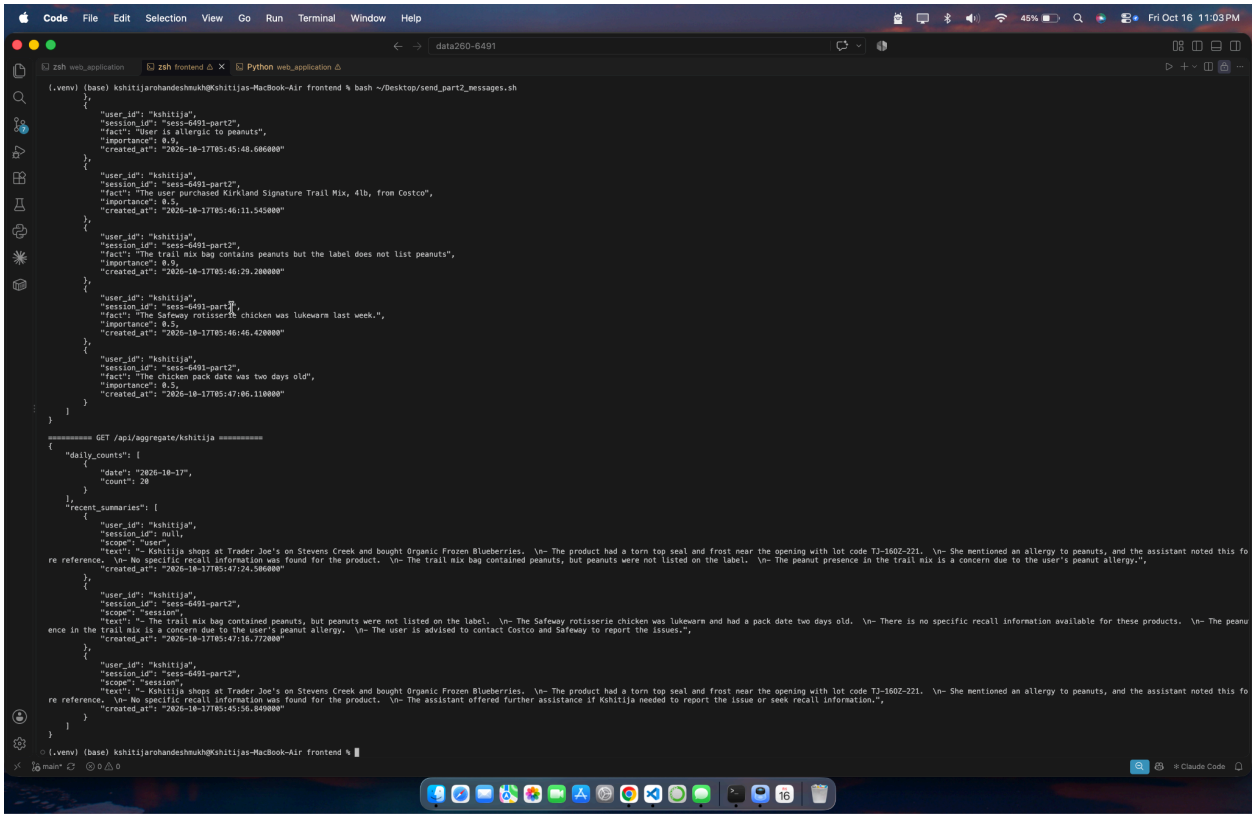
```
@router.post("/chat", response_model=ChatOut)
def post_chat(body: ChatIn):
   session_id = body.session_id or f"sess-{uuid.uuid4().hex[:8]}"
   return run_chat_turn(body.user_id, session_id, body.message.strip())

@router.get("/memory/{user_id}", response_model=MemoryOut)
def get_memory(user_id: str):
   return memory_view(user_id)

@router.get("/aggregate/{user_id}", response_model=AggregateOut)
def get_aggregate(user_id: str):
   return aggregate_view(user_id)
```



```
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air frontend % bash ~/Desktop/send_part2_messages.sh

========== USER 8 ==========
That chicken pack date was two days old.
ASSISTANT: Hello Kshitija, I'm your grocery recall assistant for project s6491. Based on the information provided:

**Long-term summaries:**
- There is no specific long-term summary available about product recalls or issues with Safeway rotisserie chicken.

**Short-term window:**
- There is no specific short-term window information available about product recalls or issues with Safeway rotisserie chicken.

**Episodic facts:**
- Last week, the Safeway rotisserie chicken from the hot case was lukewarm.
- The chicken pack date was two days old.

This is a significant concern, as the chicken should have been kept at a safe temperature and consumed within a safe timeframe. The lukewarm temperature and the pack date being two days old may indicate improper handling or storage
, which could pose a food safety risk. I strongly recommend that you do not consume the chicken and contact Safeway or the store where you purchased it to report the issue. If you have any concerns about food safety or need assista
nce with a recall, please let me know. I will help you further if needed.
short_term_count: 8
long_term_summary: - Kshitija shops at Trader Joe's on Stevens Creek and bought Organic Frozen Blueberries.
- The product had a torn top seal and frost near the opening with lot code TJ-160Z-221.
- She mentioned an allergy to peanuts, and the assistant noted this for future reference.
- No specific recall information was found for the product.
- The assistant offered further assistance if Kshitija needed to report the issue or seek recall information.
episodic_facts: [{'fact': 'The Safeway rotisserie chicken was lukewarm last week.', 'importance': 0.5}, {'fact': 'The trail mix bag contains peanuts but the label does not list peanuts', 'importance': 0.9}, {'fact': "The top seal o
f the Trader Joe's Organic Frozen Blueberries was torn.", 'importance': 0.8}]

========== USER 9 ==========
What recall issues have I already told you about?
ASSISTANT: You have already mentioned the following recall issues:

1. **Kirkland Signature Trail Mix, 4lb, from Costco** — The trail mix bag contained peanuts, but peanuts were not listed on the label. This is a concern due to your peanut allergy.

2. **Safeway rotisserie chicken** — The chicken was lukewarm and had a pack date that was two days old, which may indicate improper storage or temperature control, posing a food safety risk.

These are the recall issues you have already shared with me. If you have any more concerns or need further assistance, feel free to let me know.
short_term_count: 8
long_term_summary: - The trail mix bag contained peanuts, but peanuts were not listed on the label.
- The Safeway rotisserie chicken was lukewarm and had a pack date two days old.
- There is no specific recall information available for these products.
- The peanut presence in the trail mix is a concern due to the user's peanut allergy.
- The user is advised to contact Costco and Safeway to report the issues.
- Kshitija shops at Trader Joe's on Stevens Creek and bought Organic Frozen Blueberries.
- The product had a torn top seal and frost near the opening with lot code TJ-160Z-221.
- She mentioned an allergy to peanuts, and the assistant noted this for future reference.
- No specific recall information was found for the product.
- The trail mix bag contained peanuts, but peanuts were not listed on the label.
- The peanut presence in the trail mix is a concern due to the user's peanut allergy.
episodic_facts: [{'fact': 'The Safeway rotisserie chicken was lukewarm last week.', 'importance': 0.5}, {'fact': "Kshitija shops at Trader Joe's on Stevens Creek", 'importance': 0.5}]

========== USER 10 ==========
Which store did I say I shop at, and what am I allergic to?
ASSISTANT: You mentioned that you shop at Trader Joe's on Stevens Creek, and you are allergic to peanuts.
short_term_count: 8
long_term_summary: - The trail mix bag contained peanuts, but peanuts were not listed on the label.
- The Safeway rotisserie chicken was lukewarm and had a pack date two days old.
- There is no specific recall information available for these products.
- The peanut presence in the trail mix is a concern due to the user's peanut allergy.
- The user is advised to contact Costco and Safeway to report the issues.
- Kshitija shops at Trader Joe's on Stevens Creek and bought Organic Frozen Blueberries.
- The product had a torn top seal and frost near the opening with lot code TJ-160Z-221.
- She mentioned an allergy to peanuts, and the assistant noted this for future reference.
- No specific recall information was found for the product.
- The trail mix bag contained peanuts, but peanuts were not listed on the label.
- The peanut presence in the trail mix is a concern due to the user's peanut allergy.
```



```
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air frontend % bash ~/Desktop/send_part2_messages.sh
========== USER 10 ==========
Which store did I say I shop at, and what am I allergic to?
ASSISTANT: You mentioned that you shop at Trader Joe's on Stevens Creek, and you are allergic to peanuts.
short_term_count: 8
long_term_summary: - The trail mix bag contained peanuts, but peanuts were not listed on the label.
- The Safeway rotisserie chicken was lukewarm and had a pack date two days old.
- There is no specific recall information available for these products.
- The peanut presence in the trail mix is a concern due to the user's peanut allergy.
- The user is advised to contact Costco and Safeway to report the issues.
- Kshitija shops at Trader Joe's on Stevens Creek and bought Organic Frozen Blueberries.
- The product had a torn top seal and frost near the opening with lot code TJ-160Z-221.
- She mentioned an allergy to peanuts, and the assistant noted this for future reference.
- No specific recall information was found for the product.
- The trail mix bag contained peanuts, but peanuts were not listed on the label.
- The peanut presence in the trail mix is a concern due to the user's peanut allergy.
episodic_facts: [{'fact': 'User is allergic to peanuts', 'importance': 0.9}, {'fact': "Kshitija shops at Trader Joe's on Stevens Creek", 'importance': 0.5}, {'fact': 'The lot code was TJ-160Z-221', 'importance': 0.7}]

========== GET /api/memory/kshitija ==========
{
    "messages": [
        {
            "user_id": "kshitija",
            "session_id": "sess-6491-part2",
            "role": "user",
            "content": "The blueberries had frost near the opening. The lot code was TJ-160Z-221.",
            "created_at": "2026-10-17T05:45:26.034000"
        },
        {
            "user_id": "kshitija",
            "session_id": "sess-6491-part2",
            "role": "assistant",
            "content": "Hello Kshitija, I'm your grocery recall assistant for project s6491. Based on the information provided:\n\n**Long-term summaries:**  \n- There is no specific long-term summary available about product recalls or issues with Trader Joe's Organic Frozen
erries.\n\n**Short-term window:**  \n- There is no specific short-term window information available about product recalls or issues with Trader Joe's Organic Frozen Blueberries.\n\n**Episodic facts:**  \n- You bought Trader Joe's Organic Frozen Blueberries, 16oz, and the to
 was torn.  \n- The blueberries had frost near the opening.  \n- The lot code was TJ-160Z-221.\n\nIf you would like to report this issue or need assistance with a recall, please provide more details, and I will help you accordingly.",
            "created_at": "2026-10-17T05:45:36.879000"
        },
        {
            "user_id": "kshitija",
            "session_id": "sess-6491-part2",
            "role": "user",
            "content": "Please remember that I am allergic to peanuts.",
            "created_at": "2026-10-17T05:45:39.993000"
        },
        {
            "user_id": "kshitija",
            "session_id": "sess-6491-part2",
            "role": "assistant",
            "content": "Hello Kshitija, I'm your grocery recall assistant for project s6491. I have noted that you are allergic to peanuts. If you have any questions about product safety, recalls, or need assistance with a product that may contain peanuts, please let me kno
ill ensure to take your allergy into consideration when providing information or assistance.",
            "created_at": "2026-10-17T05:45:47.261000"
        },
        {
            "user_id": "kshitija",
            "session_id": "sess-6491-part2",
            "role": "user",
            "content": "I also bought Kirkland Signature Trail Mix, 4lb, from Costco.",
            "created_at": "2026-10-17T05:45:56.920000"
        },
        {
            "user_id": "kshitija",
            "session_id": "sess-6491-part2",
            "role": "assistant",
            "content": "Hello Kshitija, I'm your grocery recall assistant for project s6491. Based on the information provided:\n\n**Long-term summaries:**  \n- There is no specific long-term summary available about product recalls or issues with Kirkland Signature Trail Mi
, from Costco.\n\n**Short-term window:**  \n- There is no specific short-term window information available about product recalls or issues with Kirkland Signature Trail Mix, 4lb, from Costco.\n\n**Episodic facts:**  \n- You also bought Kirkland Signature Trail Mix, 4lb, fro
co.\n\nIf you would like to report this item or need assistance with a recall, please provide more details, and I will help you accordingly. I will also keep your peanut allergy in mind if there are any related concerns.",
            "created_at": "2026-10-17T05:46:09.708000"
        },
        {
            "user_id": "kshitija",
            "session_id": "sess-6491-part2",
            "role": "user",
            "content": "The trail mix bag contains peanuts but the label does not list peanuts.",
            "created_at": "2026-10-17T05:46:11.614000"
        },
        {
            "user_id": "kshitija",
```

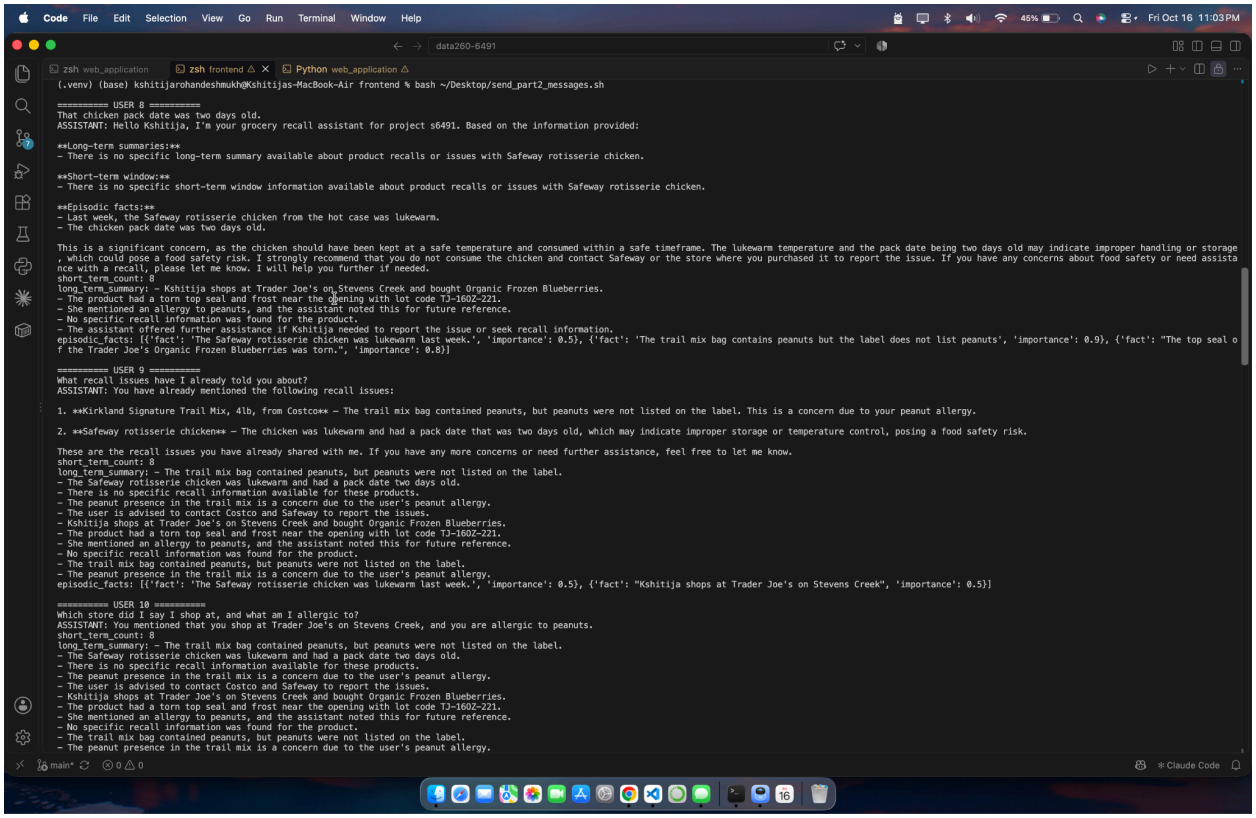

Code File Edit Selection View Go Run Terminal Window Help
Fri Oct 16 11:03 PM
data260-6491
zsh web_application
zsh frontend
Python web_application
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air frontend % bash ~/Desktop/send_part2_messages.sh
},
{
"user_id": "kshitija",
"session_id": "sess-6491-part2",
"fact": "User is allergic to peanuts",
"importance": 0.9,
"created_at": "2026-10-17T05:45:48.686000"
},
{
"user_id": "kshitija",
"session_id": "sess-6491-part2",
"fact": "The user purchased Kirkland Signature Trail Mix, 4lb, from Costco",
"importance": 0.5,
"created_at": "2026-10-17T05:46:11.545000"
},
{
"user_id": "kshitija",
"session_id": "sess-6491-part2",
"fact": "The trail mix bag contains peanuts but the label does not list peanuts",
"importance": 0.9,
"created_at": "2026-10-17T05:46:29.280000"
},
{
"user_id": "kshitija",
"session_id": "sess-6491-part2",
"fact": "The Safeway rotisserie chicken was lukewarm last week.",
"importance": 0.5,
"created_at": "2026-10-17T05:46:46.420000"
},
{
"user_id": "kshitija",
"session_id": "sess-6491-part2",
"fact": "The chicken pack date was two days old",
"importance": 0.5,
"created_at": "2026-10-17T05:47:06.110000"
}
]
}
========== GET /api/aggregate/kshitija ==========
{
"daily_counts": [
{
"date": "2026-10-17",
"count": 20
}
],
"recent_summaries": [
{
"user_id": "kshitija",
"session_id": null,
"scope": "user",
"text": "- Kshitija shops at Trader Joe's on Stevens Creek and bought Organic Frozen Blueberries. \n- The product had a torn top seal and frost near the opening with lot code TJ-160Z-221. \n- She mentioned an allergy to peanuts, and the assistant noted this fo
re reference. \n- No specific recall information was found for the product. \n- The trail mix bag contained peanuts, but peanuts were not listed on the label. \n- The peanut presence in the trail mix is a concern due to the user's peanut allergy.",
"created_at": "2026-10-17T05:47:24.586000"
},
{
"user_id": "kshitija",
"session_id": "sess-6491-part2",
"scope": "session",
"text": "- The trail mix bag contained peanuts, but peanuts were not listed on the label. \n- The Safeway rotisserie chicken was lukewarm and had a pack date two days old. \n- There is no specific recall information available for these products. \n- The peanu
ence in the trail mix is a concern due to the user's peanut allergy. \n- The user is advised to contact Costco and Safeway to report the issues.",
"created_at": "2026-10-17T05:47:16.772000"
},
{
"user_id": "kshitija",
"session_id": "sess-6491-part2",
"scope": "session",
"text": "- Kshitija shops at Trader Joe's on Stevens Creek and bought Organic Frozen Blueberries. \n- The product had a torn top seal and frost near the opening with lot code TJ-160Z-221. \n- She mentioned an allergy to peanuts, and the assistant noted this fo
re reference. \n- No specific recall information was found for the product. \n- The assistant offered further assistance if Kshitija needed to report the issue or seek recall information.",
"created_at": "2026-10-17T05:45:56.849000"
}
]
}
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air frontend %
main*
Claude Code

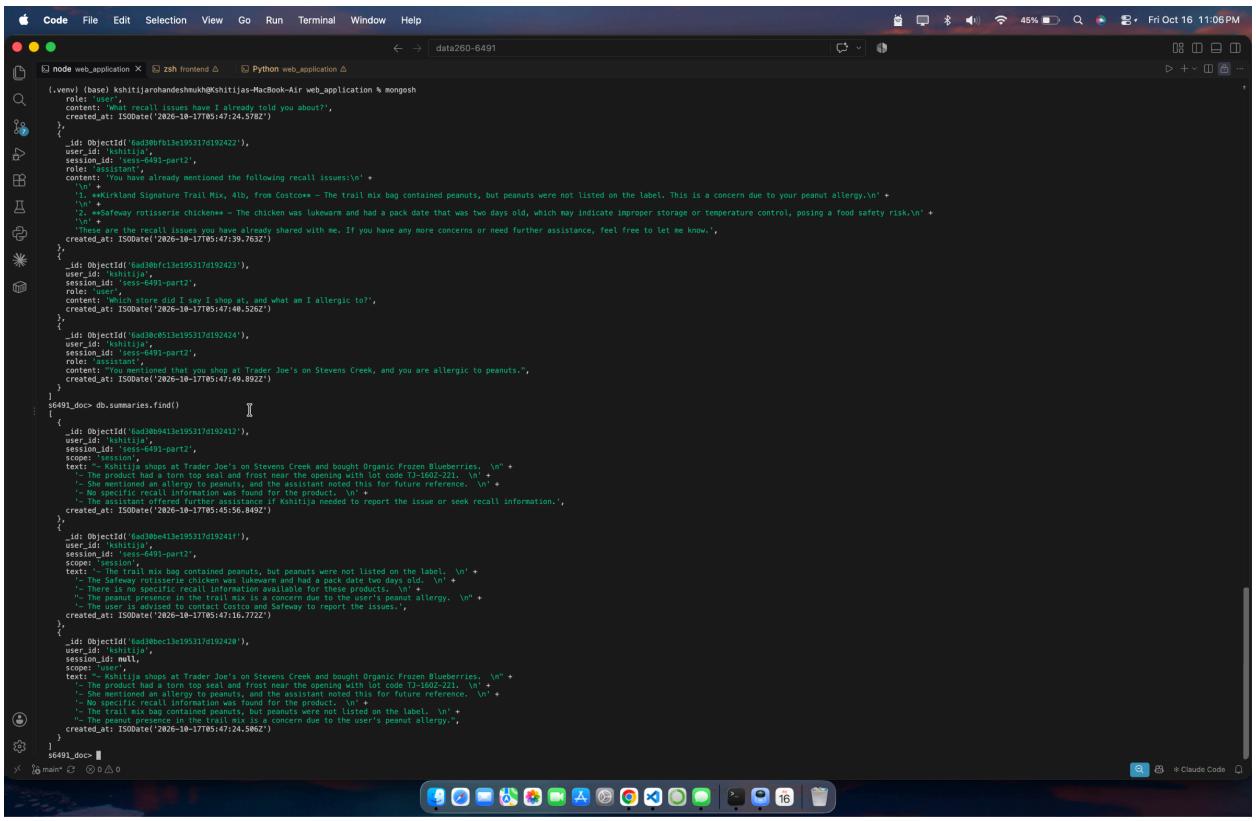
Memory logic

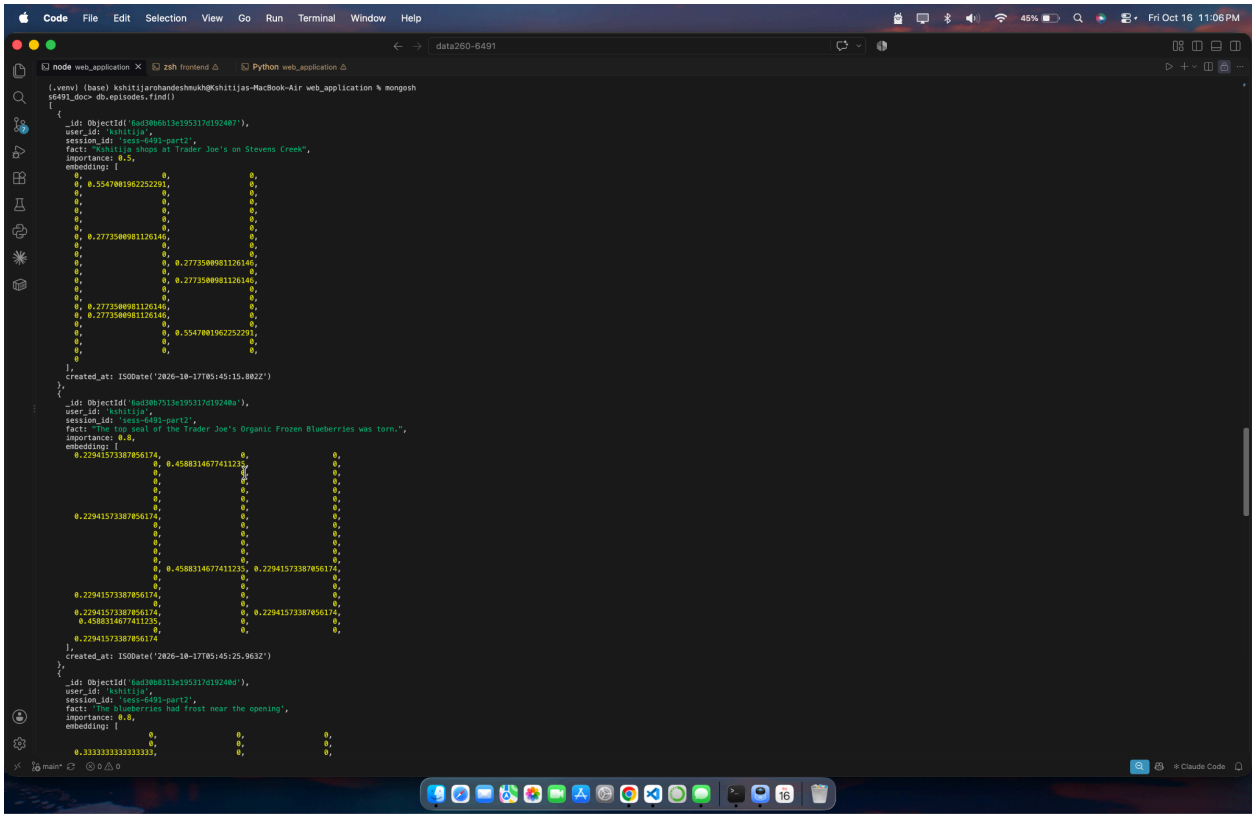
- Short-term memory: sliding window of the last N messages for the session.
- Long-term summaries: trigger every M user messages; summarize recent conversation into concise bullets under scope='session'; build/refresh a lifetime summary by condensing multiple session summaries under scope='user' with session_id = null.
- Episodic memory: extract up to 3 short facts per user message with an importance score (0..1); embed each fact and store the array in MongoDB; for each turn, retrieve the top-k relevant episodes by cosine similarity between the current message embedding and stored episodes.

```
SHORT_TERM_N = int(os.getenv("SHORT_TERM_N", "8"))
SUMMARIZE_EVERY_USER_MSGS = int(os.getenv("SUMMARIZE_EVERY_USER_MSGS", "4"))
EPISODE_TOP_K = int(os.getenv("EPISODE_TOP_K", "3"))

def load_short_term(session_id: str, limit: int) -> list:
   rows = list(get_messages().find({"session_id": session_id}).sort("created_at",
-1).limit(limit))
   rows.reverse()
   return rows

def maybe_summarize(user_id: str, session_id: str, client) -> None:
   user_turns = get_messages().count_documents({"session_id": session_id, "role":
"user"})
   if user_turns == 0 or user_turns % SUMMARIZE_EVERY_USER_MSGS != 0:
   return
# write scope="session"; every second session summary, write scope="user" with
session_id=None

def extract_facts(client, message: str) -> list:
   raw = client.complete([
   {"role": "system", "content": EXTRACT_INSTRUCTIONS},
   {"role": "user", "content": message},
   ]).content
   return parse_json_object(raw).get("facts", [])[:3]

def retrieve_episodes(user_id: str, message: str, client, top_k: int) -> list:
   query_vector = embed_text(client, message)
   ranked = []
   for episode in get_episodes().find({"user_id": user_id}):
   ranked.append((cosine(query_vector, episode.get("embedding") or []), episode))
   ranked.sort(key=lambda pair: pair[0], reverse=True)
```

```
    return [{"fact": episode["fact"], "importance": episode["importance"]} for score,
episode in ranked[:top_k] if score > 0]

system = SYSTEM_PRIMER
if long_term_lines:
    system += "\n\nLong-term memory:\n" + "\n\n".join(long_term_lines)
if fact_lines:
    system += "\n\nEpisodic facts:\n" + "\n".join(fact_lines)

messages = [{"role": "system", "content": system}]
for item in prior_messages:
    messages.append({"role": item["role"], "content": item["content"]})
messages.append({"role": "user", "content": message})
```


(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air frontend % bash ~/Desktop/send_part2_messages.sh
========= USER 8 =========
That chicken pack date was two days old.
ASSISTANT: Hello Kshitija, I'm your grocery recall assistant for project s6491. Based on the information provided:
Long-term summaries:
- There is no specific long-term summary available about product recalls or issues with Safeway rotisserie chicken.
Short-term window:
- There is no specific short-term window information available about product recalls or issues with Safeway rotisserie chicken.
Episodic facts:
- Last week, the Safeway rotisserie chicken from the hot case was lukewarm.
- The chicken pack date was two days old.
This is a significant concern, as the chicken should have been kept at a safe temperature and consumed within a safe timeframe. The lukewarm temperature and the pack date being two days old may indicate improper handling or storage , which could pose a food safety risk. I strongly recommend that you do not consume the chicken and contact Safeway or the store where you purchased it to report the issue. If you have any concerns about food safety or need assista nce with a recall, please let me know. I will help you further if needed.
short_term_count: 8
long_term_summary: - Kshitija shops at Trader Joe's on Stevens Creek and bought Organic Frozen Blueberries.
- The product had a torn top seal and frost near the opening with lot code TJ-160Z-221.
- She mentioned an allergy to peanuts, and the assistant noted this for future reference.
- No specific recall information was found for the product.
- The assistant offered further assistance if Kshitija needed to report the issue or seek recall information.
episodic_facts: [{'fact': 'The Safeway rotisserie chicken was lukewarm last week.', 'importance': 0.5}, {'fact': 'The trail mix bag contains peanuts but the label does not list peanuts', 'importance': 0.9}, {'fact': "The top seal o f the Trader Joe's Organic Frozen Blueberries was torn.", 'importance': 0.8}]
========= USER 9 =========
What recall issues have I already told you about?
ASSISTANT: You have already mentioned the following recall issues:
1. **Kirkland Signature Trail Mix, 4lb, from Costco** — The trail mix bag contained peanuts, but peanuts were not listed on the label. This is a concern due to your peanut allergy.
2. **Safeway rotisserie chicken** — The chicken was lukewarm and had a pack date that was two days old, which may indicate improper storage or temperature control, posing a food safety risk.
These are the recall issues you have already shared with me. If you have any more concerns or need further assistance, feel free to let me know.
short_term_count: 8
long_term_summary: - The trail mix bag contained peanuts, but peanuts were not listed on the label.
- The Safeway rotisserie chicken was lukewarm and had a pack date two days old.
- There is no specific recall information available for these products.
- The peanut presence in the trail mix is a concern due to the user's peanut allergy.
- The user is advised to contact Costco and Safeway to report the issues.
- Kshitija shops at Trader Joe's on Stevens Creek and bought Organic Frozen Blueberries.
- The product had a torn top seal and frost near the opening with lot code TJ-160Z-221.
- She mentioned an allergy to peanuts, and the assistant noted this for future reference.
- No specific recall information was found for the product.
- The trail mix bag contained peanuts, but peanuts were not listed on the label.
- The peanut presence in the trail mix is a concern due to the user's peanut allergy.
episodic_facts: [{'fact': 'The Safeway rotisserie chicken was lukewarm last week.', 'importance': 0.5}, {'fact': "Kshitija shops at Trader Joe's on Stevens Creek", 'importance': 0.5}]
========= USER 10 =========
Which store did I say I shop at, and what am I allergic to?
ASSISTANT: You mentioned that you shop at Trader Joe's on Stevens Creek, and you are allergic to peanuts.
short_term_count: 8
long_term_summary: - The trail mix bag contained peanuts, but peanuts were not listed on the label.
- The Safeway rotisserie chicken was lukewarm and had a pack date two days old.
- There is no specific recall information available for these products.
- The peanut presence in the trail mix is a concern due to the user's peanut allergy.
- The user is advised to contact Costco and Safeway to report the issues.
- Kshitija shops at Trader Joe's on Stevens Creek and bought Organic Frozen Blueberries.
- The product had a torn top seal and frost near the opening with lot code TJ-160Z-221.
- She mentioned an allergy to peanuts, and the assistant noted this for future reference.
- No specific recall information was found for the product.
- The trail mix bag contained peanuts, but peanuts were not listed on the label.
- The peanut presence in the trail mix is a concern due to the user's peanut allergy.


(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % mongosh
role: 'user',
content: 'What recall issues have I already told you about?',
created_at: ISODate('2026-10-17T05:47:24.578Z')
_id: ObjectId('6ad30bfb13e195317d192422'),
user_id: 'kshitija',
session_id: 'sess-6491-part2',
role: 'assistant',
content: 'You have already mentioned the following recall issues:\n' +
'\n' +
'1. **Kirkland Signature Trail Mix, 4lb, from Costco** — The trail mix bag contained peanuts, but peanuts were not listed on the label. This is a concern due to your peanut allergy.\n' +
'\n' +
'2. **Safeway rotisserie chicken** — The chicken was lukewarm and had a pack date that was two days old, which may indicate improper storage or temperature control, posing a food safety risk.\n' +
'\n' +
'These are the recall issues you have already shared with me. If you have any more concerns or need further assistance, feel free to let me know.',
created_at: ISODate('2026-10-17T05:47:39.763Z')
_id: ObjectId('6ad30bfc13e195317d192423'),
user_id: 'kshitija',
session_id: 'sess-6491-part2',
role: 'user',
content: 'Which store did I say I shop at, and what am I allergic to?',
created_at: ISODate('2026-10-17T05:47:40.526Z')
_id: ObjectId('6ad30c0513e195317d192424'),
user_id: 'kshitija',
session_id: 'sess-6491-part2',
role: 'assistant',
content: "You mentioned that you shop at Trader Joe's on Stevens Creek, and you are allergic to peanuts.",
created_at: ISODate('2026-10-17T05:47:49.892Z')
s6491_doc> db.summaries.find()
_id: ObjectId('6ad30b9413e195317d192412'),
user_id: 'kshitija',
session_id: 'sess-6491-part2',
scope: 'session',
text: '- Kshitija shops at Trader Joe's on Stevens Creek and bought Organic Frozen Blueberries. \n' +
'- The product had a torn top seal and frost near the opening with lot code TJ-160Z-221. \n' +
'- She mentioned an allergy to peanuts, and the assistant noted this for future reference. \n' +
'- No specific recall information was found for the product. \n' +
'- The assistant offered further assistance if Kshitija needed to report the issue or seek recall information.',
created_at: ISODate('2026-10-17T05:45:56.849Z')
_id: ObjectId('6ad30be413e195317d19241f'),
user_id: 'kshitija',
session_id: 'sess-6491-part2',
scope: 'session',
text: '- The trail mix bag contained peanuts, but peanuts were not listed on the label. \n' +
'- The Safeway rotisserie chicken was lukewarm and had a pack date two days old. \n' +
'- There is no specific recall information available for these products. \n' +
'- The peanut presence in the trail mix is a concern due to the user's peanut allergy. \n' +
'- The user is advised to contact Costco and Safeway to report the issues.',
created_at: ISODate('2026-10-17T05:47:16.772Z')
_id: ObjectId('6ad30bec13e195317d192420'),
user_id: 'kshitija',
session_id: null,
scope: 'user',
text: '- Kshitija shops at Trader Joe's on Stevens Creek and bought Organic Frozen Blueberries. \n' +
'- The product had a torn top seal and frost near the opening with lot code TJ-160Z-221. \n' +
'- She mentioned an allergy to peanuts, and the assistant noted this for future reference. \n' +
'- No specific recall information was found for the product. \n' +
'- The trail mix bag contained peanuts, but peanuts were not listed on the label. \n' +
'- The peanut presence in the trail mix is a concern due to the user's peanut allergy.',
created_at: ISODate('2026-10-17T05:47:24.506Z')
s6491_doc>


Code File Edit Selection View Go Run Terminal Window Help
Fri Oct 16 11:06 PM
data260-6491
node web_application
zsh frontend
Python web_application
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air web_application % mongosh
s6491_doc> db.episodes.find()
_id: ObjectId('6ad30b6b13e195317d192407'),
user_id: 'kshitija',
session_id: 'sess-6491-part2',
fact: "Kshitija shops at Trader Joe's on Stevens Creek",
importance: 0.5,
_id: ObjectId('6ad30b7513e195317d19240a'),
fact: "The top seal of the Trader Joe's Organic Frozen Blueberries was torn.",
importance: 0.8,
_id: ObjectId('6ad30b8313e195317d19240d'),
fact: 'The blueberries had frost near the opening',
importance: 0.8,
created_at: ISODate('2026-10-17T05:45:15.802Z')
created_at: ISODate('2026-10-17T05:45:25.963Z')
main*
Claude Code

How to run and test
Send at least 10 messages in one session (via Postman/cURL) so the short-term window, summary generation, and episodic retrieval can all be observed. Optionally build a simple HTML page with one input box calling POST /api/chat to simulate a mini chat interface - not required, but recommended for screenshots.

Submit: a conversation screenshot showing the assistant's replies; the JSON result of GET /api/memory/{user_id}; the JSON result of GET /api/aggregate/{user_id}; MongoDB Compass showing the messages, summaries, and episodes collections with documents; and your code for the three endpoints plus the memory logic functions (episode extraction, summarization trigger, embedding + retrieval, prompt assembly).

Done as per above screenshots.

Part 3 - What Memory Costs

1. **Log memory usage**

For each /api/chat request, record:

- Prompt and completion tokens
- Tokens used by the system primer, long-term summary, short-term window, and episodic facts
- Request latency

Save the raw results as CSV or JSON.

```
log_chat_turn({
   "prompt_tokens": result.usage.input_tokens,
   "completion_tokens": result.usage.output_tokens,
   "primer_tokens": estimate_tokens(SYSTEM_PRIMER),
   "long_term_tokens": estimate_tokens(long_term),
   "short_term_tokens": estimate_tokens(short_term_text),
   "episodic_tokens": estimate_tokens(episodic_text),
   "latency_ms": int((time.perf_counter() - started) * 1000),
})
```

2. **Create memory probes**

Before testing, create memory_probes.yaml with five fixed questions, expected answer criteria, and a 1–5 scoring rubric.

```
rubric:
 5: All expected facts are present and specific.
 1: None of the expected facts are present.
probes:
 - id: name_store
   question: What is my name, and which store location do I shop at?
   expected: ["Kshitija", "Stevens Creek"]
```

3. **Compare short-term memory sizes**

Run the same 30-message conversation in fresh sessions with:

- SHORT_TERM_N = 4
- SHORT_TERM_N = 12
- SHORT_TERM_N = 30

Ask the same five probe questions after each run. Report the mean prompt tokens per turn, mean latency, and mean probe-quality score.

4. **Test one memory-saving policy**

Implement one eviction or compression policy. Repeat the conversation and probes, then report token savings and any change in probe quality.

3 and 4 ->

```
def short_term_n() -> int:
    return int(os.getenv("SHORT_TERM_N", "8"))


def apply_memory_policy(prior: list, episodes: list) -> tuple:
    if memory_policy() != "compress":
        return prior, episodes
    if len(prior) > 4:
        older = " | ".join(item["content"].replace("\n", " ")[:40] for item in
prior[:-4])
        prior = [{"role": "user", "content": "Compressed earlier turns: " + older}] +
prior[-4:]
    return prior, episodes[:1]
```

5. **Estimate memory cost**

Using input pricing of $1 per million tokens and output pricing of $5 per million tokens, estimate the cost of 10,000 30-turn conversations under your best and worst settings.

```
def cost_for_10000(prompt_mean: float, completion_mean: float) -> float:
    turns = 30 * 10000
    return (prompt_mean * turns) / 1_000_000 * 1 + (completion_mean * turns) /
1_000_000 * 5
```

No paid hosted-model run is required. **Explain how you selected the best and worst settings. Submit the memory-cost table, memory_probes.yaml, and the eviction/compression comparison results.**

Setting	Mean prompt tokens/turn	Mean latency (ms)	Probe quality (1 - 5)
SHORTTERMN=4	614.8	17577	3.4
SHORTTERMN=12	979.2	26304	4.6
SHORTTERMN=30	1572	32529	4.2
Eviction/compression policy	774.5	27085	4.2

Best setting: SHORT_TERM_N=12.
Worst setting: SHORT_TERM_N=30.

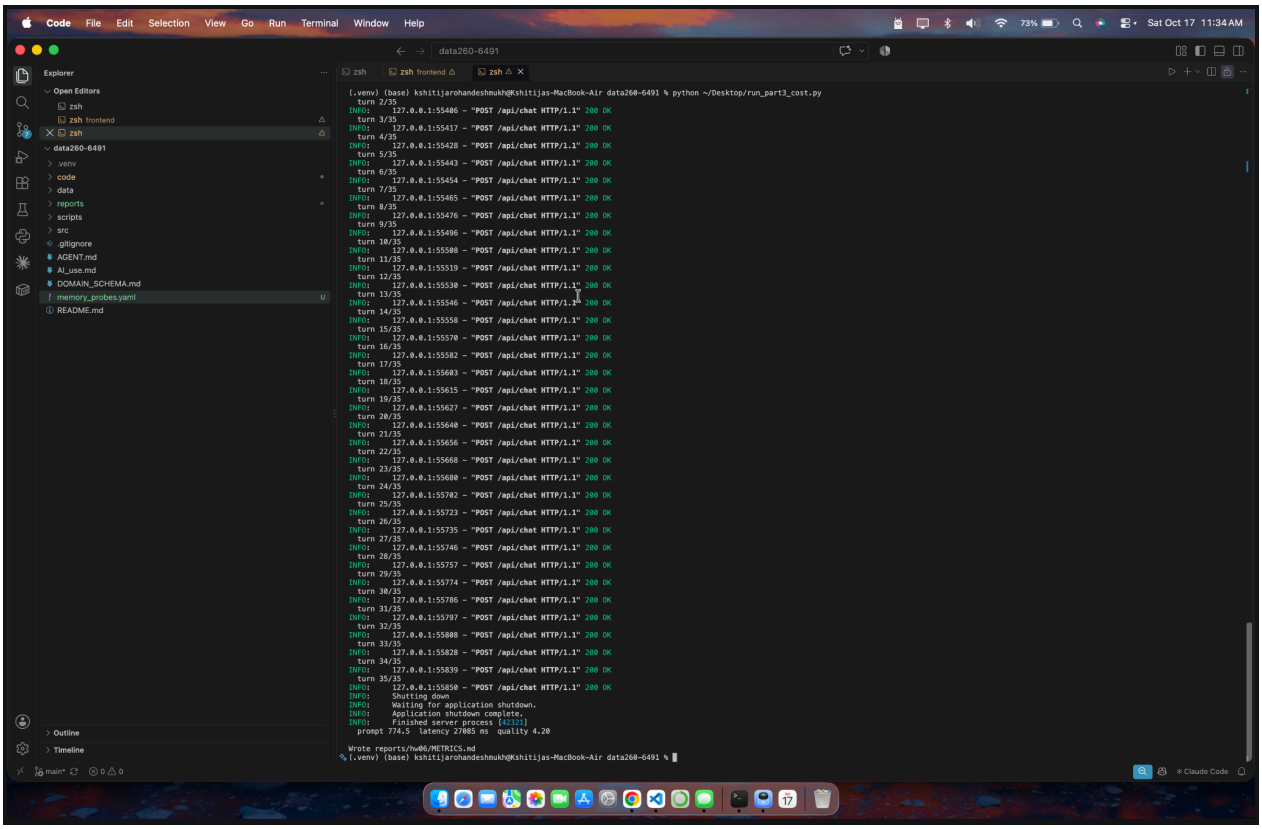
The best setting is the one with the highest probe score. If two scores are close, the lower prompt-token setting is preferred. The worst setting is the one with the highest mean prompt tokens.

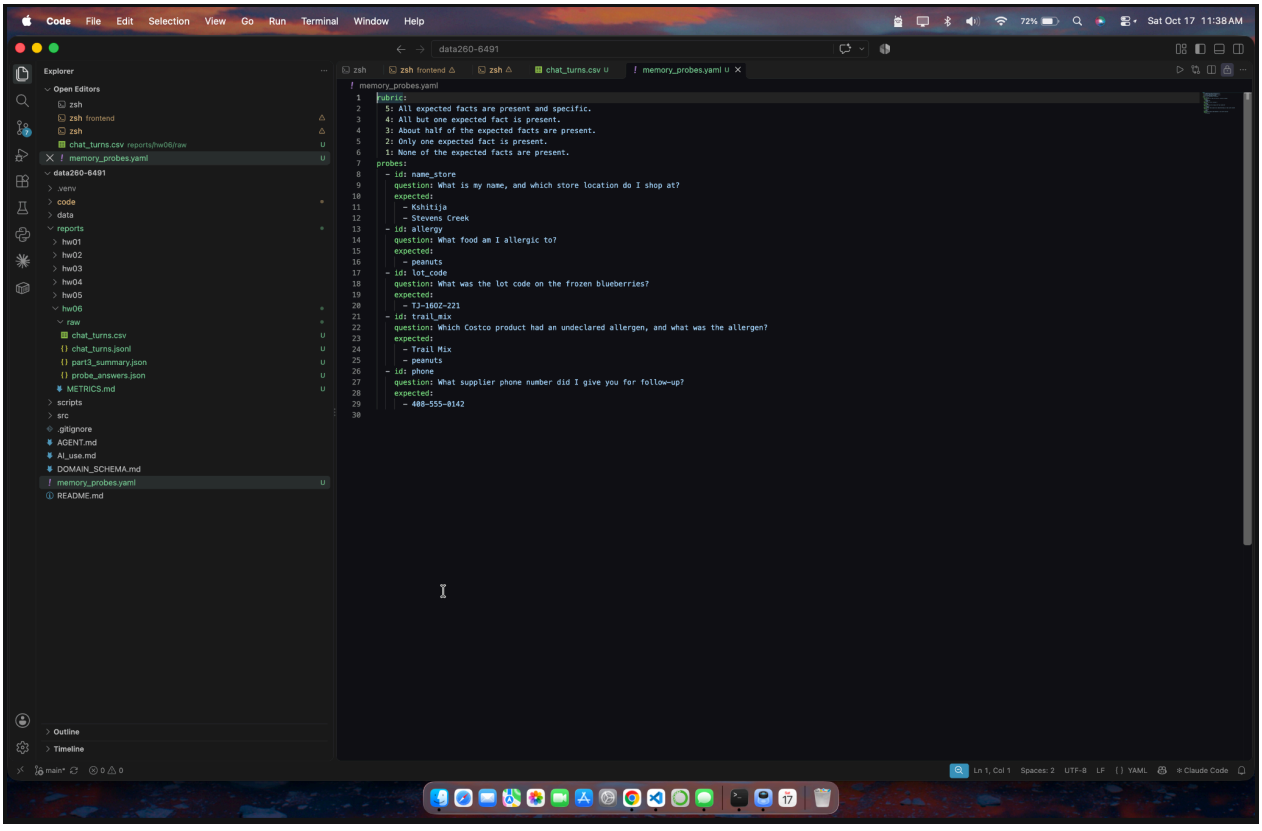
Cost of 10,000 conversations of 30 turns, at $1 per million input tokens and $5 per million output tokens:

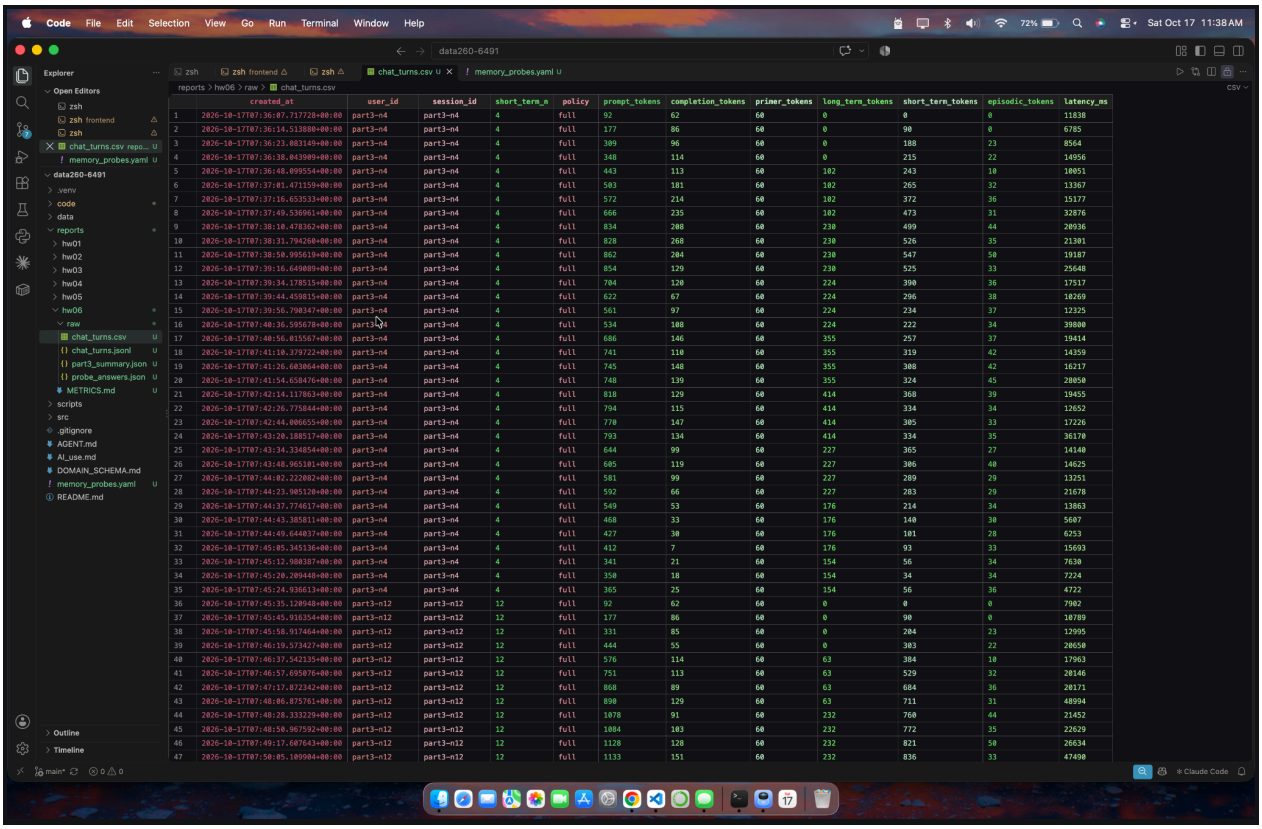
- Best (SHORT_TERM_N=12): $445.02
- Worst (SHORT_TERM_N=30): $602.49

Prompt and completion token means come from Ollama. Primer, summary, short-term, and episodic token columns in chat_turns.csv are estimates of 4 characters per token, because Ollama returns only the total prompt size.

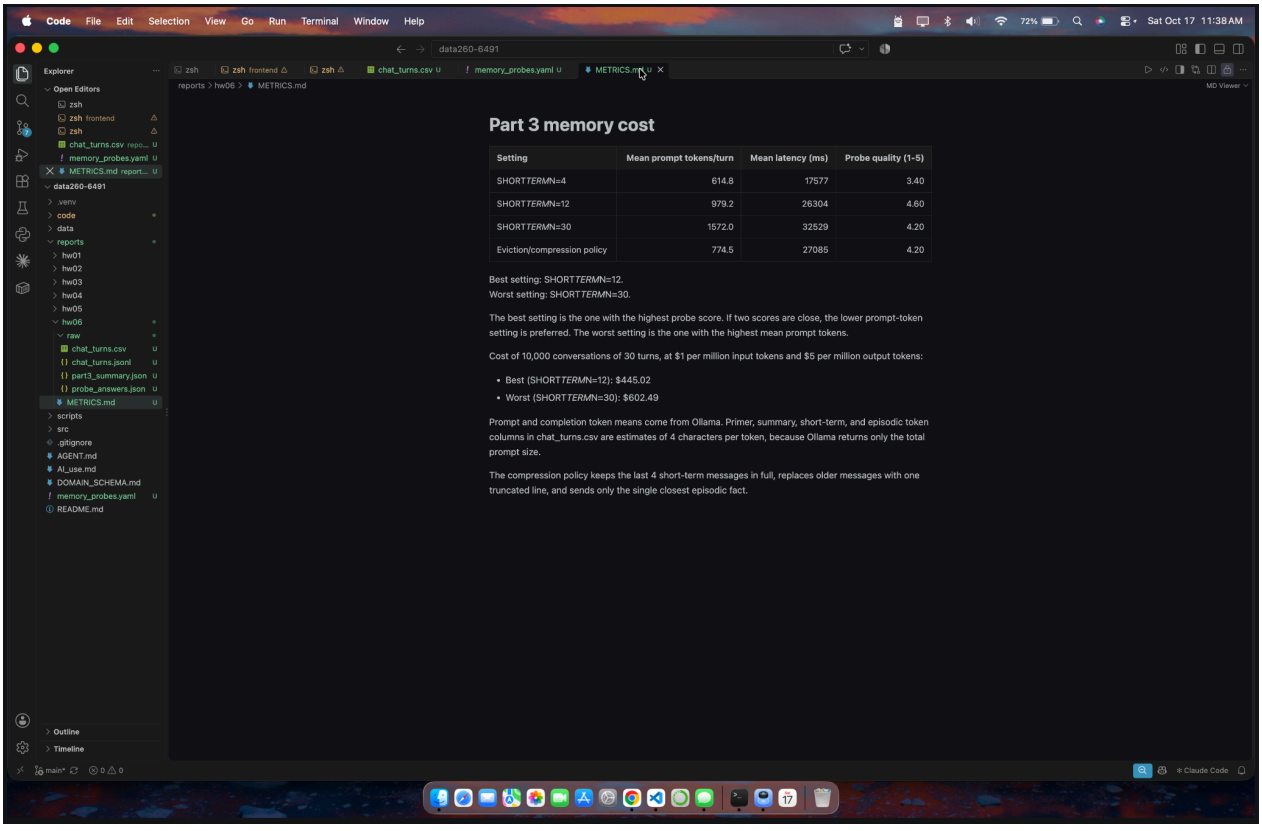
The compression policy keeps the last 4 short-term messages in full, replaces older messages with one truncated line, and sends only the single closest episodic fact.


(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air data260-6491 % python ~/Desktop/run_part3_cost.py
turn 2/35
INFO: 127.0.0.1:55406 - "POST /api/chat HTTP/1.1" 200 OK
turn 3/35
INFO: 127.0.0.1:55417 - "POST /api/chat HTTP/1.1" 200 OK
turn 4/35
INFO: 127.0.0.1:55428 - "POST /api/chat HTTP/1.1" 200 OK
turn 5/35
INFO: 127.0.0.1:55443 - "POST /api/chat HTTP/1.1" 200 OK
turn 6/35
INFO: 127.0.0.1:55454 - "POST /api/chat HTTP/1.1" 200 OK
turn 7/35
INFO: 127.0.0.1:55465 - "POST /api/chat HTTP/1.1" 200 OK
turn 8/35
INFO: 127.0.0.1:55476 - "POST /api/chat HTTP/1.1" 200 OK
turn 9/35
INFO: 127.0.0.1:55496 - "POST /api/chat HTTP/1.1" 200 OK
turn 10/35
INFO: 127.0.0.1:55508 - "POST /api/chat HTTP/1.1" 200 OK
turn 11/35
INFO: 127.0.0.1:55519 - "POST /api/chat HTTP/1.1" 200 OK
turn 12/35
INFO: 127.0.0.1:55530 - "POST /api/chat HTTP/1.1" 200 OK
turn 13/35
INFO: 127.0.0.1:55546 - "POST /api/chat HTTP/1.1" 200 OK
turn 14/35
INFO: 127.0.0.1:55558 - "POST /api/chat HTTP/1.1" 200 OK
turn 15/35
INFO: 127.0.0.1:55570 - "POST /api/chat HTTP/1.1" 200 OK
turn 16/35
INFO: 127.0.0.1:55582 - "POST /api/chat HTTP/1.1" 200 OK
turn 17/35
INFO: 127.0.0.1:55603 - "POST /api/chat HTTP/1.1" 200 OK
turn 18/35
INFO: 127.0.0.1:55615 - "POST /api/chat HTTP/1.1" 200 OK
turn 19/35
INFO: 127.0.0.1:55627 - "POST /api/chat HTTP/1.1" 200 OK
turn 20/35
INFO: 127.0.0.1:55640 - "POST /api/chat HTTP/1.1" 200 OK
turn 21/35
INFO: 127.0.0.1:55656 - "POST /api/chat HTTP/1.1" 200 OK
turn 22/35
INFO: 127.0.0.1:55668 - "POST /api/chat HTTP/1.1" 200 OK
turn 23/35
INFO: 127.0.0.1:55680 - "POST /api/chat HTTP/1.1" 200 OK
turn 24/35
INFO: 127.0.0.1:55702 - "POST /api/chat HTTP/1.1" 200 OK
turn 25/35
INFO: 127.0.0.1:55723 - "POST /api/chat HTTP/1.1" 200 OK
turn 26/35
INFO: 127.0.0.1:55735 - "POST /api/chat HTTP/1.1" 200 OK
turn 27/35
INFO: 127.0.0.1:55746 - "POST /api/chat HTTP/1.1" 200 OK
turn 28/35
INFO: 127.0.0.1:55757 - "POST /api/chat HTTP/1.1" 200 OK
turn 29/35
INFO: 127.0.0.1:55774 - "POST /api/chat HTTP/1.1" 200 OK
turn 30/35
INFO: 127.0.0.1:55786 - "POST /api/chat HTTP/1.1" 200 OK
turn 31/35
INFO: 127.0.0.1:55797 - "POST /api/chat HTTP/1.1" 200 OK
turn 32/35
INFO: 127.0.0.1:55808 - "POST /api/chat HTTP/1.1" 200 OK
turn 33/35
INFO: 127.0.0.1:55828 - "POST /api/chat HTTP/1.1" 200 OK
turn 34/35
INFO: 127.0.0.1:55839 - "POST /api/chat HTTP/1.1" 200 OK
turn 35/35
INFO: 127.0.0.1:55850 - "POST /api/chat HTTP/1.1" 200 OK
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [42321]
prompt 774.5 latency 27085 ms quality 4.20
Wrote reports/hw06/METRICS.md
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air data260-6491 %


rubric:
5: All expected facts are present and specific.
4: All but one expected fact is present.
3: About half of the expected facts are present.
2: Only one expected fact is present.
1: None of the expected facts are present.
probes:
- id: name_store
question: What is my name, and which store location do I shop at?
expected:
- Kshitija
- Stevens Creek
- id: allergy
question: What food am I allergic to?
expected:
- peanuts
- id: lot_code
question: What was the lot code on the frozen blueberries?
expected:
- TJ-16OZ-221
- id: trail_mix
question: Which Costco product had an undeclared allergen, and what was the allergen?
expected:
- Trail Mix
- peanuts
- id: phone
question: What supplier phone number did I give you for follow-up?
expected:
- 408-555-0142



	created_at	user_id	session_id	short_term_n	policy	prompt_tokens	completion_tokens	primer_tokens	long_term_tokens	short_term_tokens	episodic_tokens	latency_ms
1	2026-10-17T07:36:07.717728+00:00	part3-n4	part3-n4	4	full	92	62	60	0	0	0	11838
2	2026-10-17T07:36:14.513888+00:00	part3-n4	part3-n4	4	full	177	86	60	0	90	0	6785
3	2026-10-17T07:36:23.083149+00:00	part3-n4	part3-n4	4	full	309	96	60	0	188	23	8564
4	2026-10-17T07:36:38.043909+00:00	part3-n4	part3-n4	4	full	348	114	60	0	215	22	14956
5	2026-10-17T07:36:48.099554+00:00	part3-n4	part3-n4	4	full	443	113	60	102	243	10	10051
6	2026-10-17T07:37:01.471159+00:00	part3-n4	part3-n4	4	full	503	101	60	102	265	32	13367
7	2026-10-17T07:37:16.653533+00:00	part3-n4	part3-n4	4	full	572	214	60	102	372	36	15177
8	2026-10-17T07:37:49.536961+00:00	part3-n4	part3-n4	4	full	666	235	60	102	473	31	32876
9	2026-10-17T07:38:10.478362+00:00	part3-n4	part3-n4	4	full	834	208	60	230	499	44	20936
10	2026-10-17T07:38:31.794260+00:00	part3-n4	part3-n4	4	full	828	268	60	230	526	35	21301
11	2026-10-17T07:38:50.995619+00:00	part3-n4	part3-n4	4	full	862	204	60	230	547	50	19187
12	2026-10-17T07:39:16.649089+00:00	part3-n4	part3-n4	4	full	854	129	60	230	525	33	25648
13	2026-10-17T07:39:34.178515+00:00	part3-n4	part3-n4	4	full	704	120	60	224	390	36	17517
14	2026-10-17T07:39:44.459815+00:00	part3-n4	part3-n4	4	full	622	67	60	224	296	38	10269
15	2026-10-17T07:39:56.798347+00:00	part3-n4	part3-n4	4	full	561	97	60	224	234	37	12325
16	2026-10-17T07:40:36.595678+00:00	part3-n4	part3-n4	4	full	534	108	60	224	222	34	39880
17	2026-10-17T07:40:56.015567+00:00	part3-n4	part3-n4	4	full	686	146	60	355	257	37	19414
18	2026-10-17T07:41:10.379722+00:00	part3-n4	part3-n4	4	full	741	110	60	355	319	42	14359
19	2026-10-17T07:41:26.603064+00:00	part3-n4	part3-n4	4	full	745	148	60	355	308	42	16217
20	2026-10-17T07:41:54.658476+00:00	part3-n4	part3-n4	4	full	748	139	60	355	324	45	20050
21	2026-10-17T07:42:14.117863+00:00	part3-n4	part3-n4	4	full	818	129	60	414	368	39	19455
22	2026-10-17T07:42:26.775844+00:00	part3-n4	part3-n4	4	full	794	115	60	414	334	34	12652
23	2026-10-17T07:42:44.006655+00:00	part3-n4	part3-n4	4	full	770	147	60	414	305	33	17226
24	2026-10-17T07:43:20.188517+00:00	part3-n4	part3-n4	4	full	793	134	60	414	334	35	36170
25	2026-10-17T07:43:34.334854+00:00	part3-n4	part3-n4	4	full	644	99	60	227	365	27	14140
26	2026-10-17T07:43:48.965101+00:00	part3-n4	part3-n4	4	full	605	119	60	227	306	40	14625
27	2026-10-17T07:44:02.222082+00:00	part3-n4	part3-n4	4	full	581	99	60	227	289	29	13251
28	2026-10-17T07:44:23.905120+00:00	part3-n4	part3-n4	4	full	592	66	60	227	283	29	21678
29	2026-10-17T07:44:37.774617+00:00	part3-n4	part3-n4	4	full	549	53	60	176	214	34	13863
30	2026-10-17T07:44:43.385811+00:00	part3-n4	part3-n4	4	full	468	33	60	176	140	30	5607
31	2026-10-17T07:44:49.644037+00:00	part3-n4	part3-n4	4	full	427	30	60	176	101	28	6253
32	2026-10-17T07:45:05.345136+00:00	part3-n4	part3-n4	4	full	412	7	60	176	93	33	15693
33	2026-10-17T07:45:12.980387+00:00	part3-n4	part3-n4	4	full	341	21	60	154	56	34	7630
34	2026-10-17T07:45:20.209448+00:00	part3-n4	part3-n4	4	full	350	18	60	154	34	34	7224
35	2026-10-17T07:45:24.936613+00:00	part3-n4	part3-n4	4	full	365	25	60	154	56	36	4722
36	2026-10-17T07:45:35.120948+00:00	part3-n12	part3-n12	12	full	92	62	60	0	0	0	7902
37	2026-10-17T07:45:45.916354+00:00	part3-n12	part3-n12	12	full	177	86	60	0	90	0	10789
38	2026-10-17T07:45:58.917464+00:00	part3-n12	part3-n12	12	full	331	85	60	0	204	23	12995
39	2026-10-17T07:46:19.573427+00:00	part3-n12	part3-n12	12	full	444	55	60	0	303	22	20650
40	2026-10-17T07:46:37.542135+00:00	part3-n12	part3-n12	12	full	576	114	60	63	384	10	17963
41	2026-10-17T07:46:57.695076+00:00	part3-n12	part3-n12	12	full	751	113	60	63	529	32	20146
42	2026-10-17T07:47:17.872342+00:00	part3-n12	part3-n12	12	full	868	89	60	63	684	36	20171
43	2026-10-17T07:48:06.875761+00:00	part3-n12	part3-n12	12	full	890	129	60	63	711	31	48994
44	2026-10-17T07:48:28.333229+00:00	part3-n12	part3-n12	12	full	1078	91	60	232	760	44	21452
45	2026-10-17T07:48:50.967592+00:00	part3-n12	part3-n12	12	full	1084	103	60	232	772	35	22629
46	2026-10-17T07:49:17.607643+00:00	part3-n12	part3-n12	12	full	1128	128	60	232	821	50	26634
47	2026-10-17T07:50:05.109904+00:00	part3-n12	part3-n12	12	full	1133	151	60	232	836	33	47490



Part 3 memory cost

Setting	Mean prompt tokens/turn	Mean latency (ms)	Probe quality (1-5)
SHORT*TERM*N=4	614.8	17577	3.40
SHORT*TERM*N=12	979.2	26304	4.60
SHORT*TERM*N=30	1572.0	32529	4.20
Eviction/compression policy	774.5	27085	4.20

Best setting: SHORT*TERM*N=12.
Worst setting: SHORT*TERM*N=30.

The best setting is the one with the highest probe score. If two scores are close, the lower prompt-token setting is preferred. The worst setting is the one with the highest mean prompt tokens.

Cost of 10,000 conversations of 30 turns, at $1 per million input tokens and $5 per million output tokens:

- Best (SHORT*TERM*N=12): $445.02
- Worst (SHORT*TERM*N=30): $602.49

Prompt and completion token means come from Ollama. Primer, summary, short-term, and episodic token columns in chat_turns.csv are estimates of 4 characters per token, because Ollama returns only the total prompt size.

The compression policy keeps the last 4 short-term messages in full, replaces older messages with one truncated line, and sends only the single closest episodic fact.

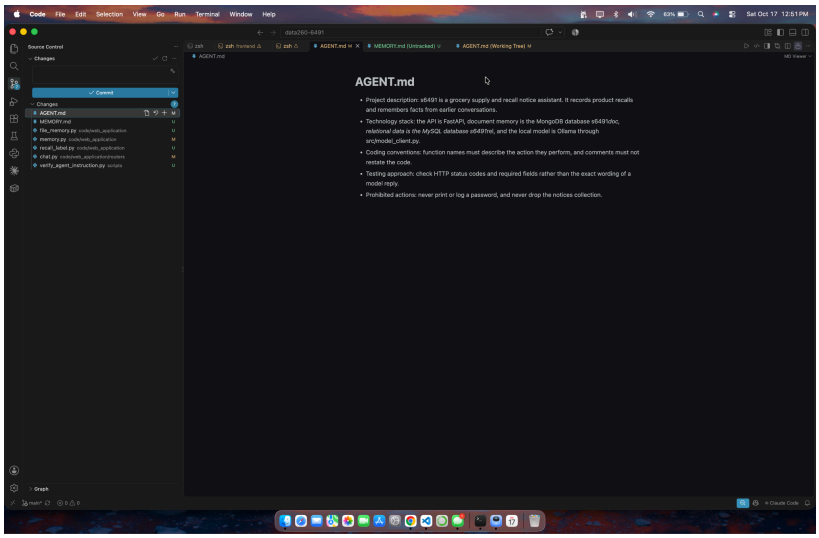
Part 4 - File-Based Memory and Consolidation

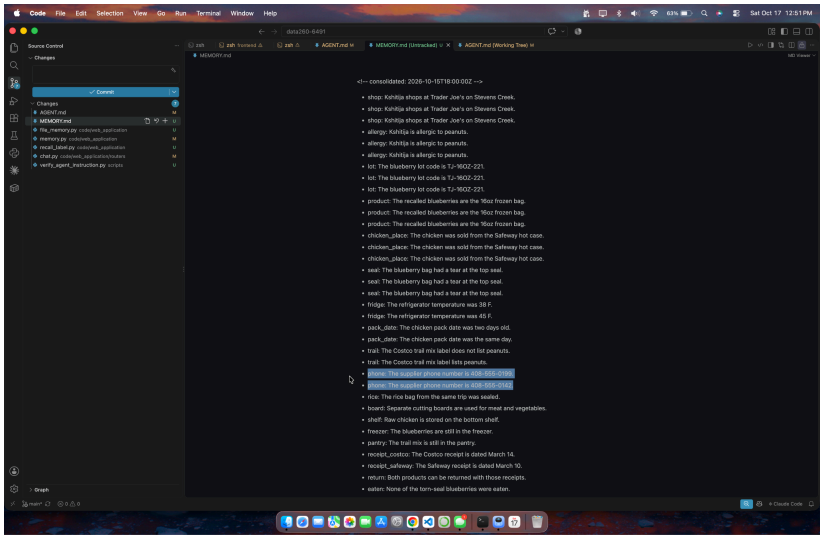
In this part, you will build a second memory system using AGENT.md and MEMORY.md, then compare it with your MongoDB-based memory system.

1. Create the memory files

Reuse and expand the AGENT.md file from HW1.

- AGENT.md stores durable project and agent instructions.
- MEMORY.md stores durable facts learned from previous conversations.





2. Implement file-based memory

Create functions that:

- Load both Markdown files
- Add their contents to the system prompt
- Read the last-consolidated timestamp
- Determine when memory consolidation is required

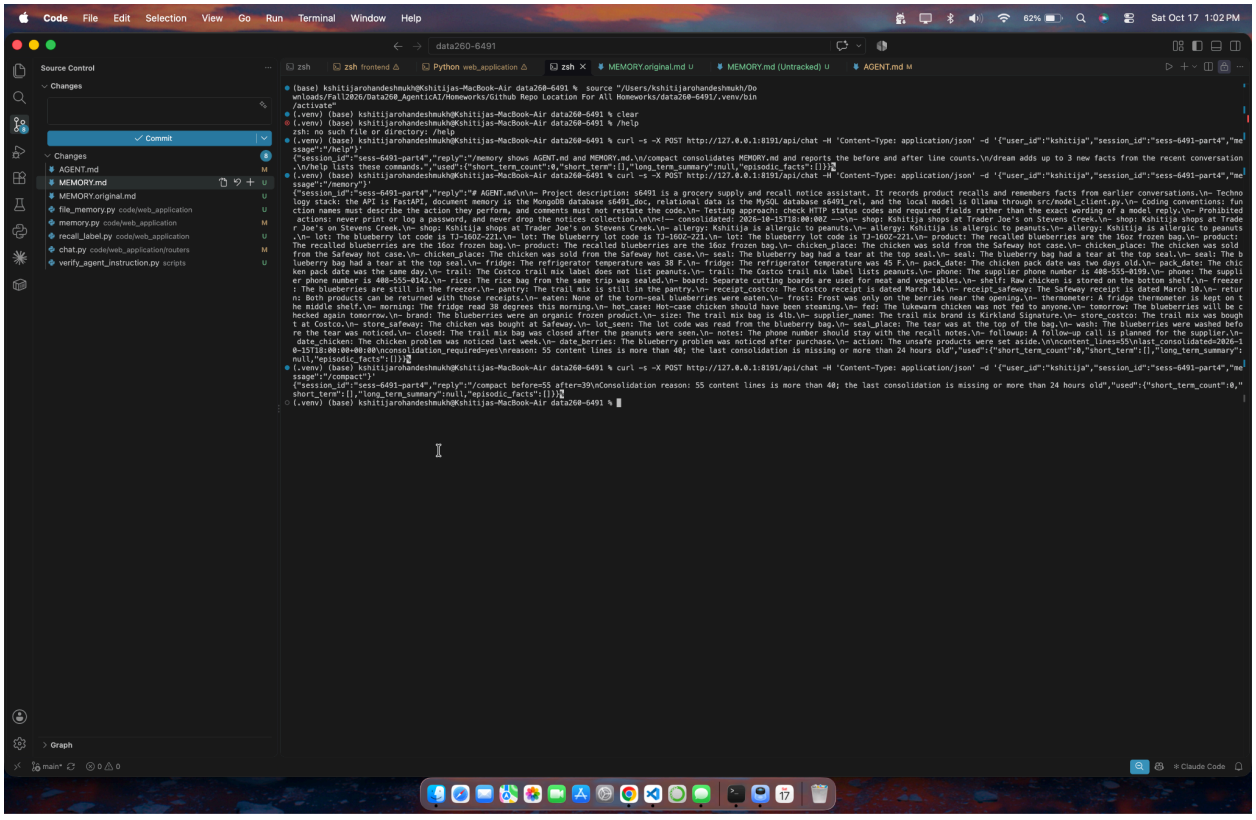
Consolidate MEMORY.md when:

- It contains more than 40 non-empty, non-comment lines, or
- Its recorded consolidation time is more than 24 hours old

```
def file_memory_block() -> str:
    agent = read_text(agent_path()).strip()
    memory = read_text(memory_path()).strip()
    return f"File memory instructions:\n{agent}\n\nDurable facts:\n{memory}"
```

```
def consolidation_required(text: str) -> tuple:
    count = len(content_lines(text))
    recorded = last_consolidated_at(text)
    too_long = count > LINE_LIMIT
    too_old = recorded is None or utcnow() - recorded > STALE_AFTER
```

```
    from file_memory import file_memory_block


    system = SYSTEM_PRIMER + "\n\n" + file_memory_block()
```


Code File Edit Selection View Go Run Terminal Window Help
Sat Oct 17 1:02 PM
data260-6491
Source Control
Changes
Commit
AGENT.md
MEMORY.md
MEMORY.original.md
file_memory.py code/web_application
memory.py code/web_application
recall_label.py code/web_application
chat.py code/web_application/routers
verify_agent_instruction.py scripts
Graph
Claude Code

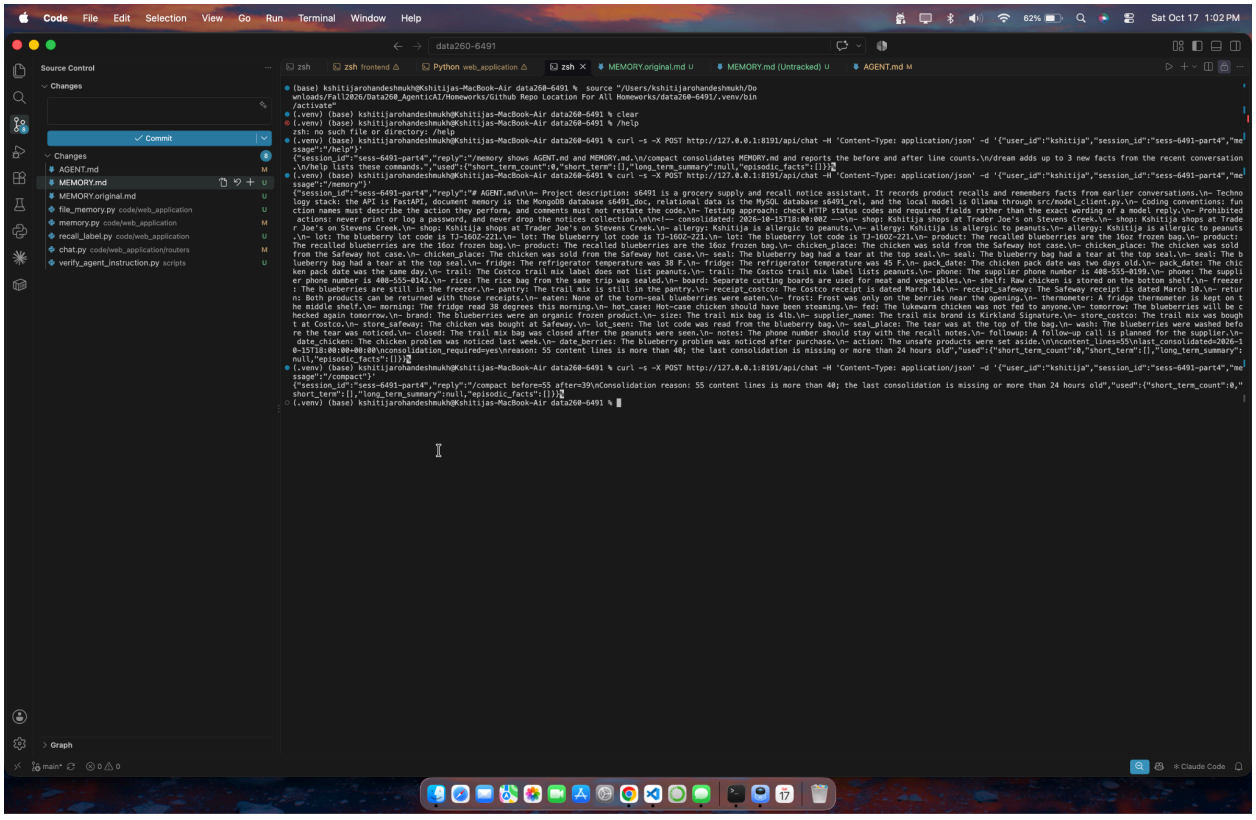
3. Support memory commands

Implement the following commands:

- /memory — display the current file-based memory
- /compact — consolidate MEMORY.md and report the message count before and after
- /dream — identify useful long-term facts from recent conversation and add them to MEMORY.md
- /help — explain the available commands

```python
def post_chat(body: ChatIn):
    message = body.message.strip()
    session_id = body.session_id or f"sess-{uuid.uuid4().hex[:8]}"
    if message.startswith("/"):
        return run_memory_command(message, body.user_id, body.session_id or "")
```

```python
def run_memory_command(command: str, user_id: str, session_id: str) -> dict:
    name = command.strip().split()[0]
    if name == "/help":
        ...
    elif name == "/memory":
        ...
    elif name == "/compact":
        reply = compact_memory()
    elif name == "/dream":
        reply = remember_recent_facts(user_id, session_id)
```

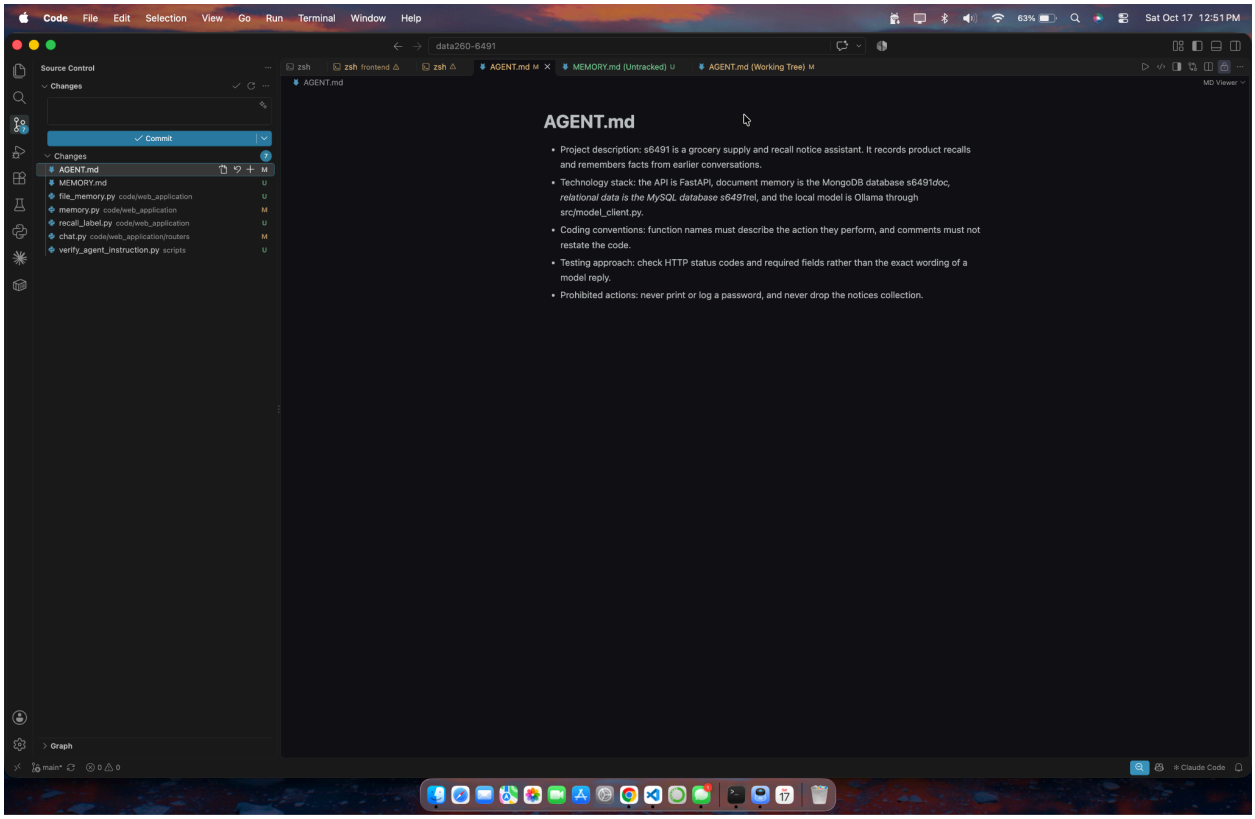

Code File Edit Selection View Go Run Terminal Window Help
Sat Oct 17 1:02 PM
data260-6491
Source Control
Changes
Commit
AGENT.md
MEMORY.md
MEMORY.original.md
file_memory.py code/web_application
memory.py code/web_application
recall_label.py code/web_application
chat.py code/web_application/routers
verify_agent_instruction.py scripts
Graph
zsh
zsh frontend
Python web_application
MEMORY.original.md
MEMORY.md (Untracked)
AGENT.md
Claude Code

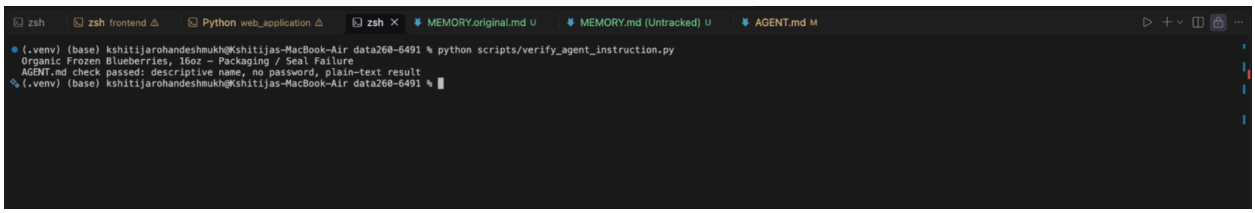
4. Expand and test AGENT.md

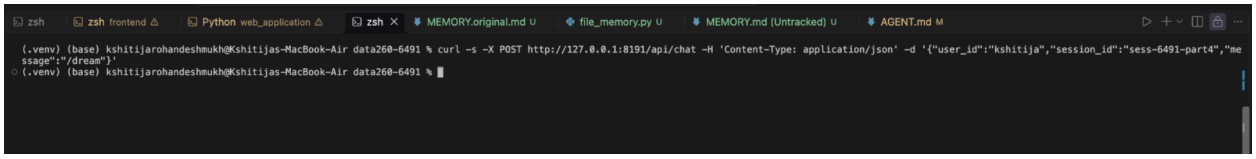
By the end of HW6, AGENT.md must contain at least five instructions covering at least three of these categories:

- Project description
- Technology stack
- Coding conventions
- Testing approach
- Prohibited actions Generate or implement one function and verify that it follows the instructions in AGENT.md.

```python
def format_recall_notice(product_name: str, recall_type: str) -> str:
    return f"{product_name} — {recall_type}"
```






zsh
zsh frontend
Python web_application
zsh
MEMORY.original.md U
file_memory.py U
MEMORY.md (Untracked) U
AGENT.md M
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air data260-6491 % curl -s -X POST http://127.0.0.1:8191/api/chat -H 'Content-Type: application/json' -d '{"user_id":"kshitija","session_id":"sess-6491-part4","message":"/dream"}'
(.venv) (base) kshitijarohandeshmukh@Kshitijas-MacBook-Air data260-6491 %

5. Test consolidation

Create a 55-line MEMORY.md containing:

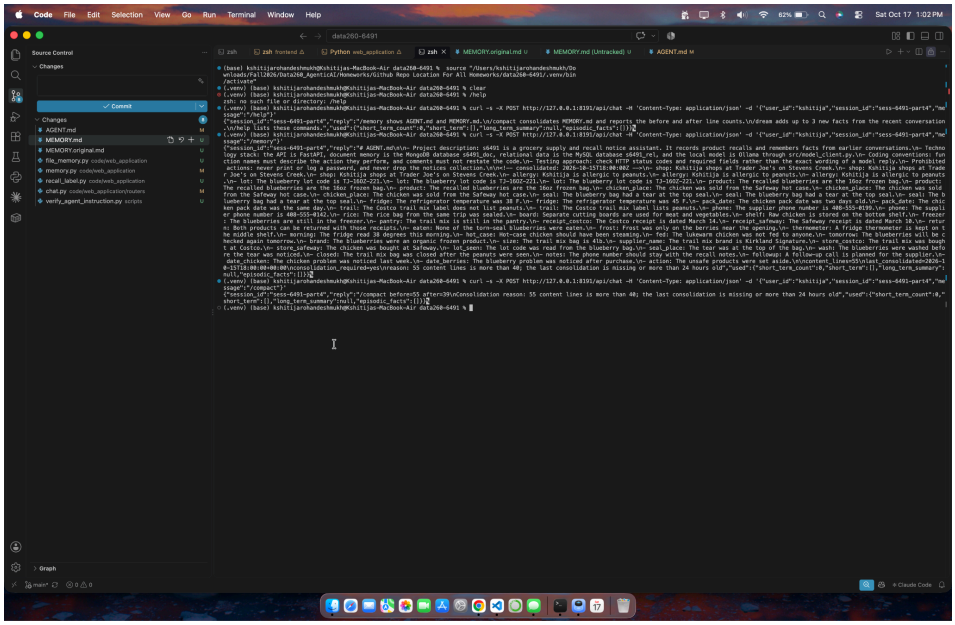
- At least three duplicated facts
- At least three deliberately contradictory facts

Run your consolidation process and reduce the file to fewer than 40 lines.
Your consolidation must:

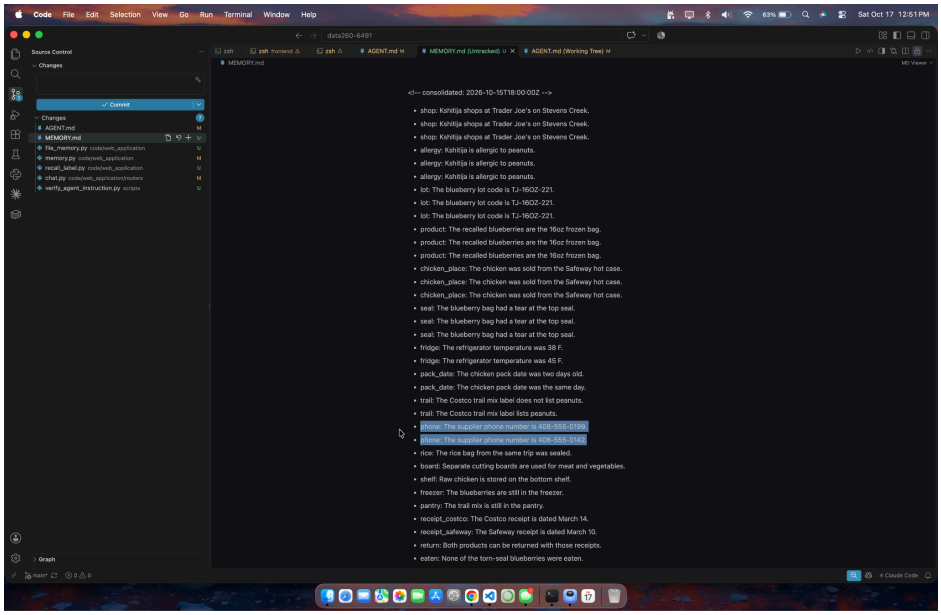
- Merge duplicate facts
- Resolve contradictions explicitly
- Avoid silently deleting conflicting information
- Identify any contradiction that was resolved incorrectly Commit both the original and consolidated versions to Git.

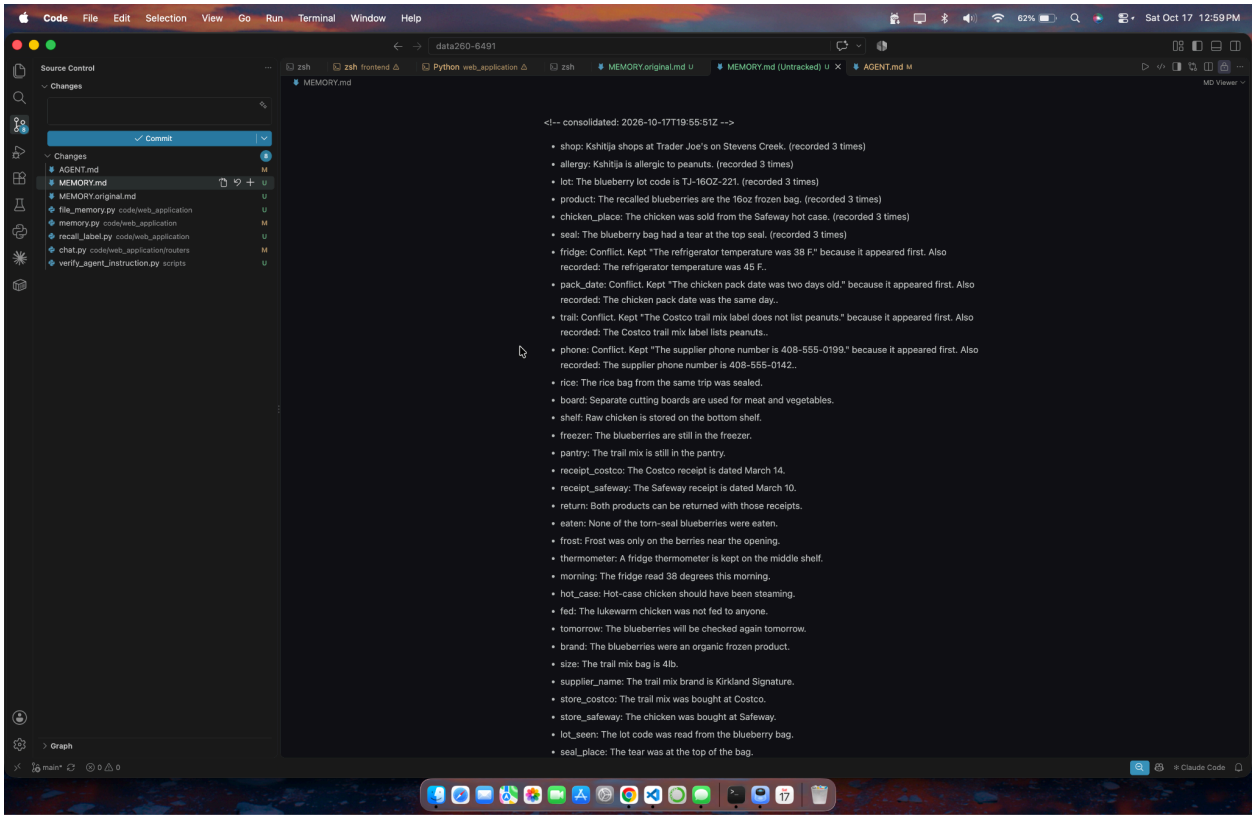
The /compact command reduced MEMORY.md from 55 content lines to 39 content lines (before=55, after=39).

```
        if len(unique) == 1:
            suffix = f" (recorded {len(values)} times)" if len(values) > 1 else ""
            written.append(f"- {key}: {unique[0]}{suffix}")
        else:
            rest = "; ".join(unique[1:])
            written.append(
                f"- {key}: Conflict. Kept \"{unique[0]}\" because it appeared first.
Also recorded: {rest}."
            )
```



The phone contradiction was resolved incorrectly. Consolidation kept 408-555-0199 because it appeared first. The correct number is 408-555-0142. The other number is still written on the same line.



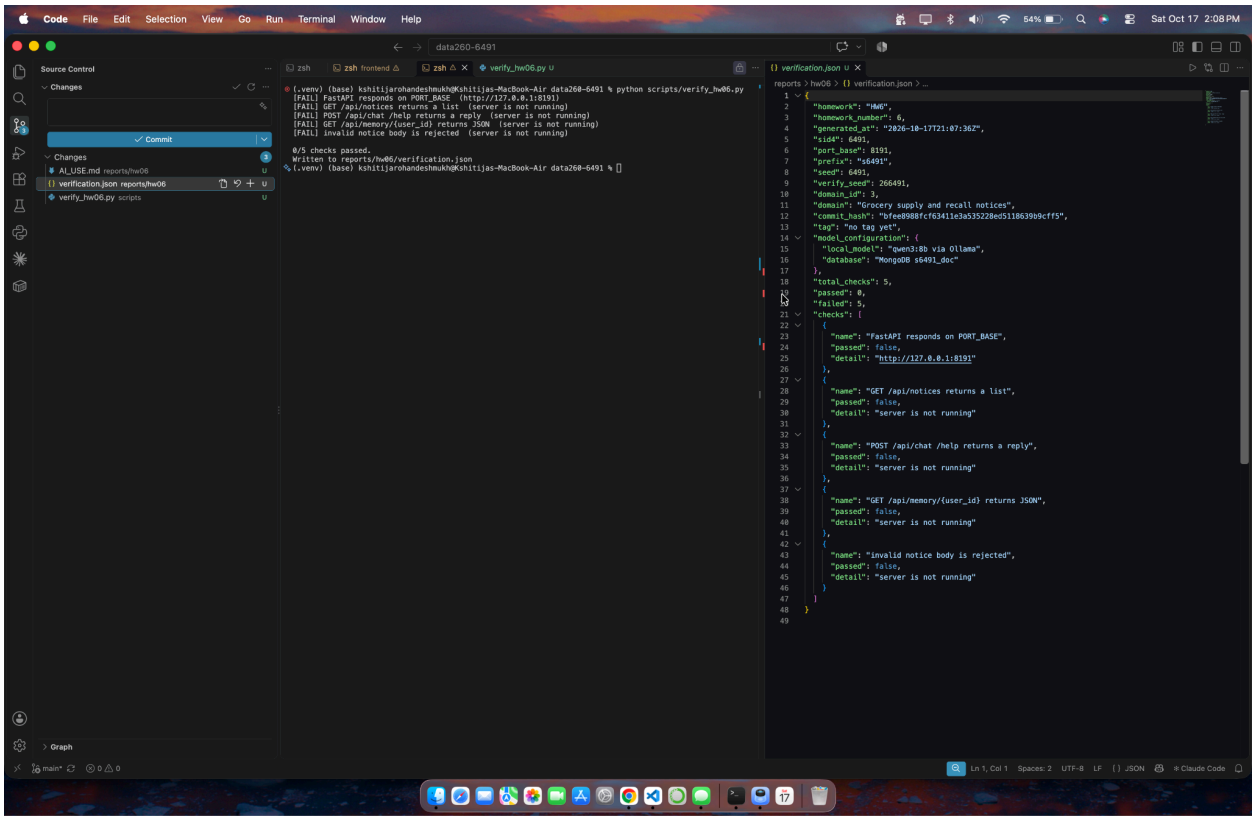


6. Compare both memory systems

Compare MongoDB-backed memory and file-based memory in terms of:

- Inspectability
- Persistence after a restart
- Scalability across many sessions
- Suitability for facts that must not be forgotten

1. MongoDB memory is useful when there are many users and conversations. We can inspect it by searching the messages, summaries, and episodes stored for each user or session. File-based memory is easier to inspect because we can simply open MEMORY.md and read it.
2. Both systems keep their data after the application restarts. MongoDB stores its data in the “s6491_doc” database, while file-based memory stores information directly on the computer.
3. MongoDB is better for handling many sessions because each user's data is stored separately. A single Markdown file can become large and difficult to organize.
4. File-based memory is better for important facts and instructions that should always be remembered. MongoDB may store a fact but not include it in a response if that fact is not considered relevant to the current conversation.



AI_USE.md

1. What did you use an AI assistant for, and what did you do yourself?

I used an AI assistant when I got stuck on the memory parts. It helped me draft the MongoDB, the file-based commands, the consolidation logic. It also helped me to resolved the errors I got.

What I did myself: I set up the repo, set up MongoDB, and kept the FastAPI server and Ollama running. I sent the curl requests, looked at the collections in mongosh, and took the screenshots. I filled the Part 3 table from METRICS.md and checked the report against the assignment.

2. One AI-produced output that was wrong or unsuitable

The server command I was given was uvicorn code.web_application.main:app while I was already inside the web_application folder. It looked like the normal start command, but the server never came up.

3. How I detected the problem

I ran it and read the traceback. It ended with ModuleNotFoundError: No module named 'code.web_application'; 'code' is not a package. Python was picking up its own code module instead of my project folder, so the app never started.

4. What was changed and why it works now

From the web_application folder I started it with uvicorn main:app --host 127.0.0.1 --port 8191. The server stayed up after that, and the notice and chat requests started returning normal HTTP responses.